BIOL 230 MIDTERM 2 REVIEW SHEET

11. Introduction to Traditional Ecological Knowledge

Summary of Robin Wall Kimmerer, ESA Recent Advances Lecture Speaker, August 15, 2019

A. Big picture:

- **Thesis**: Solving ecological crises needs symbiosis between Western science (p-values) and Indigenous knowledge (cultural values).
- **Problem**: We face a crisis of values, not just a data deficit; more facts alone rarely change behavior.
- **Goal**: Decolonize ecology → move from an intellectual monoculture to knowledge diversity and authentic inclusion.

B. Key Ideas

- **Two-Eyed Seeing**: Use both Western analytic and Indigenous relational lenses.
- **Knowledge Diversity**: Intellectual monoculture = weak culture; diversity fuels innovation.
- **Decolonizing Ecology**: Include not only Indigenous people but Indigenous ideas.
- **Knowledge + Responsibility:** Knowing implies obligation to care for land and life.

C. Indigenous Perspectives

- Humans are part of nature, not separate observers.
- Land is kin, teacher, sacred, not property or "natural resource."
- TEK (Traditional Ecological Knowledge) is long-term, place-based, holistic (mind, body, emotion, spirit).
- Five R's: Respect, Relationship, Responsibility, Reciprocity, Reverence.

D. Fortress vs Garden

- Fortress Science
    - Built from reductionist "bricks."

        - Each **brick** = one **scientific fact**, **dataset**, or **paper** built through reductionist methods (breaking nature into parts to study them separately).
    - Excludes emotion/spirit.
    - Commodifies knowledge.
    - Erases Indigenous contributions.
- Garden of Knowledge (Three Sisters Metaphor)
    - Corn = Indigenous Knowledge: ethical framework, holistic worldview.
    - Beans = Western Science: curiosity, analytical strength—needs guidance of "Corn".
    - Squash = Institutional Climate: protects and nurtures cooperation.
    - Together, they yield greater productivity and sustainability than alone.

E. What Authentic Inclusion Means

- Beyond representation → include other ways of knowing.
- Create "ethical spaces" where both knowledge systems guide research.
- Reimagine criteria of "good science" (e.g., reciprocity, equity, humility).
- Support Indigenous sovereignty, language, and biocultural landscapes.

F. Sustainability Reframed

- Western view: "*How can we keep taking sustainably?*"
- Indigenous view: "*What can we give back?*"
- People can be beneficial actors—right harvesting and care increase biodiversity.

G. Practice & policy implications

- Research criteria (reimagined): Does a project show respect for humans/non-human persons? Support reciprocity? Acknowledge community knowledge holders? Share benefits equitably? Recognize land as teacher?

- Education: Territorial acknowledgements, language revitalization support, biocultural landscape protection, curricula that integrate TEK at the center, not as folklore sidebars.
- Careers/mentoring: Model the "both-and"—publish well and engage communities; change norms so early-career folks aren't penalized.

H. Stories/examples to cite

- Goldenrod & asters (beauty question dismissed as "not science" → example of narrowed epistemology).
- Fire as good for land "rediscovered" by science though long known in TEK.
- Menominee Forest: Indigenous-led forestry sustaining both economy and old-growth structure.

I. Fast glossary

- Traditional Ecological Knowledge (TEK): Cumulative, place-based, relational knowledge tied to duties and ethics.
- Decolonizing ecology: Undoing erasure; integrating multiple knowledge systems on equal footing; shifting power/benefit structures.
- Scientism: Belief that science is the only valid way of knowing.
- Biocultural diversity: Co-occurrence and co-dependence of cultural and biological diversity.

12. Introduction to communities and diversity

A. Key concepts

- Communities are groups of interacting species that occur in the same place at the same time
- Species diversity and species composition are important descriptors of community structure

- Communities are often defined for convenience, e.g., by environment or habitat type, depending on the ecological question(s) of interest:
    - Desert, Hot springs, Coral reef

B. How do we define a "community"?

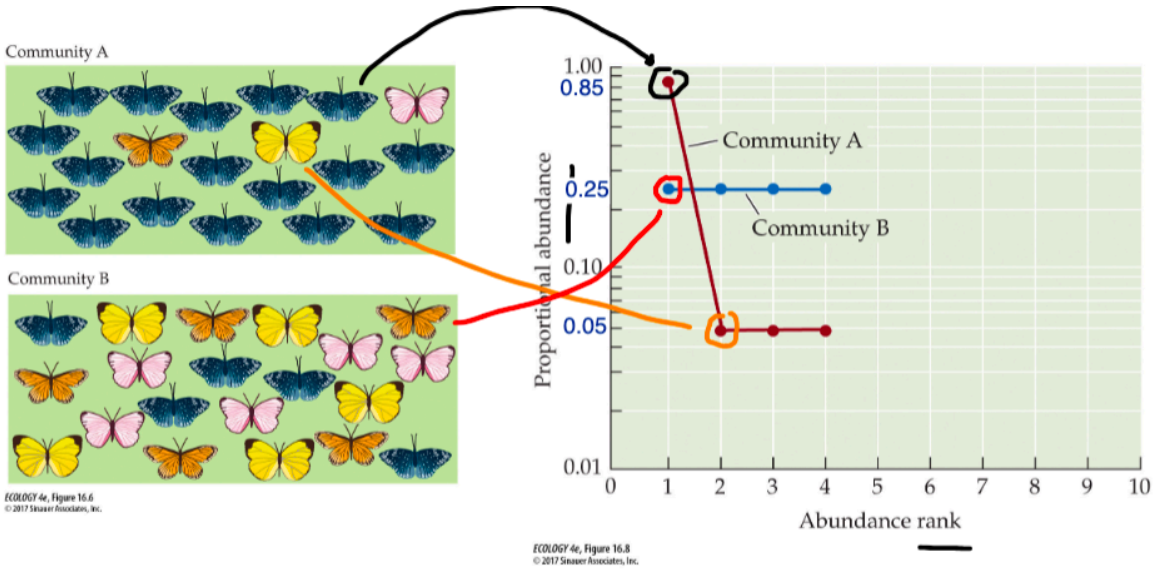
- taxonomic affinity (e.g., the avian community)
- Guild (species that use the same resource, e.g. flower feeders)
- Functional group (species that function in similar ways, e.g, nitrogen fixing plants)

C. Community structure

- Definition: the set of characteristics that shape communities.
- Key elements:
    - species composition (the identity of the species present in the community)
    - species' ecological inter-relationships
    - species diversity

D. Species diversity vs. biodiversity

- Species diversity combines two aspects of a community: species richness and species evenness.
    - Species richness—the number of species in a community.
    - Species evenness—relative abundances compared with one another
- Species evenness can be visualized with rank abundance curves



- Sloper deeper = Lower evenness = less diversity

E. Shannon index: Diversity index

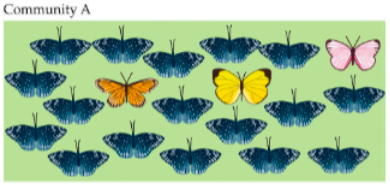
$$H = -\sum_{i=1}^{S} p_i \ln(p_i)$$

p_i = proportion of individuals in the ith species

S = number of species in the community

- Calculating the Shannon Index of species diversity:

Community A



Community B



TABLE 16.1 Calculation of Species Diversity Using the Shannon Index for Communities A and B

COMMUNITY A

SPECIES	ABUNDANCE	PROPORTION (p_i)	ln (p_i)	p_i ln (p_i)
Blue	17	0.85	−0.163	−0.139
Yellow	1	0.05	−2.996	−0.150
Pink	1	0.05	−2.996	−0.150
Orange	1	0.05	−2.996	−0.150
Total	20	1.00		−0.589

COMMUNITY B

SPECIES	ABUNDANCE	PROPORTION (p_i)	ln (p_i)	p_i ln (p_i)
Blue	5	0.25	−1.386	−0.347
Yellow	5	0.25	−1.386	−0.347
Pink	5	0.25	−1.386	−0.347
Orange	5	0.25	−1.386	−0.347
Total	20	1.00		−1.388

$$H = -\sum_{i=1}^{s} p_i \ln(p_i)$$

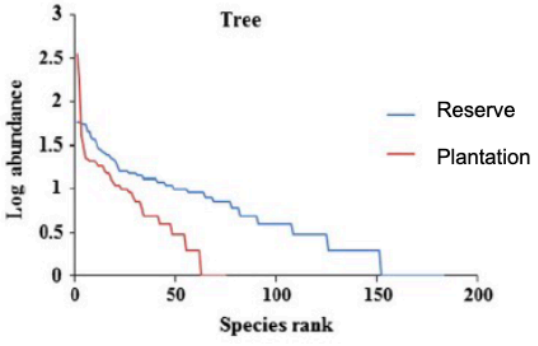
Community A: $H = 0.589$

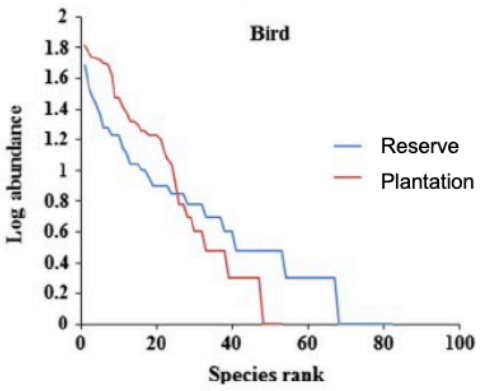
Community B: $H = 1.388$

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- Higher Shannon Index (H) = Greater diversity
- evenness J can be calculated from H and S
  - $J = H / \ln(S)$

Comparing diversity patterns in forest reserves vs. plantations ("private forest") in Malaysia





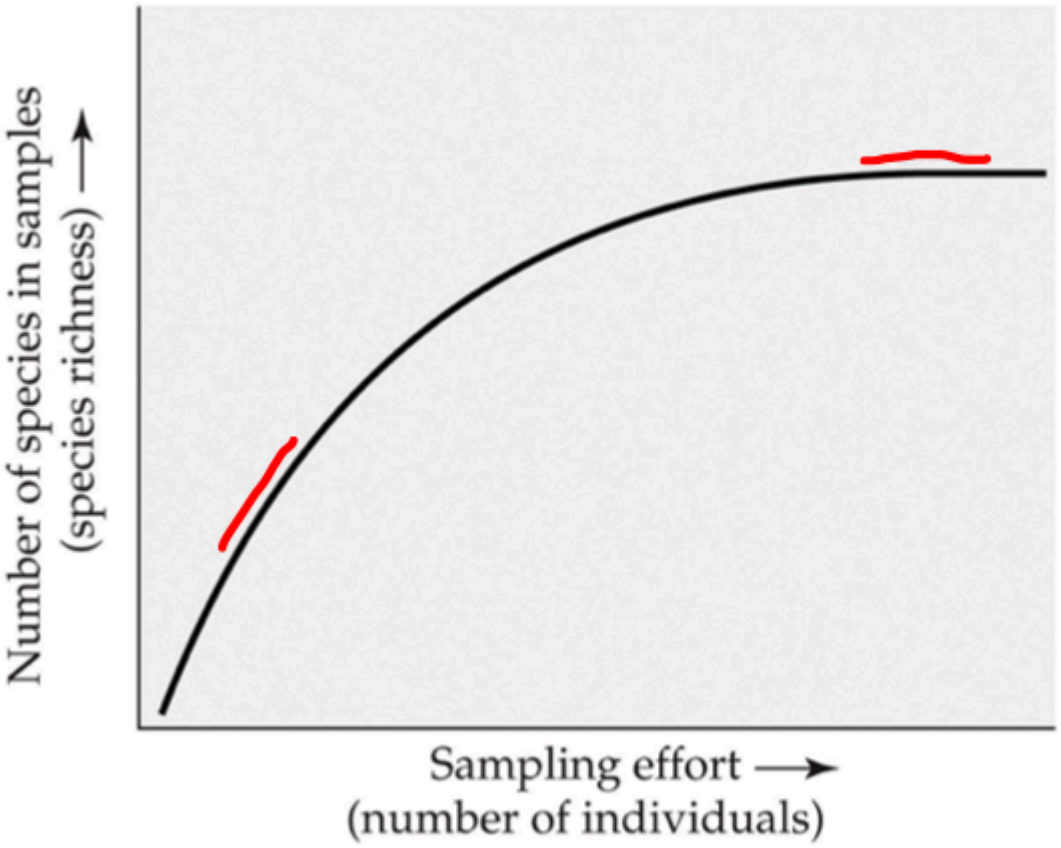
- The higher right-hand endpoints (183 for trees, 82 for birds) for the blue lines indicates higher richness in the reserve
- The lower slopes of the blue lines indicates higher evenness in the reserve
- Because richness and evenness are both higher in the reserve, Shannon diversity is also higher in the reserve

13. Communities and diversity continued

A. Key concepts

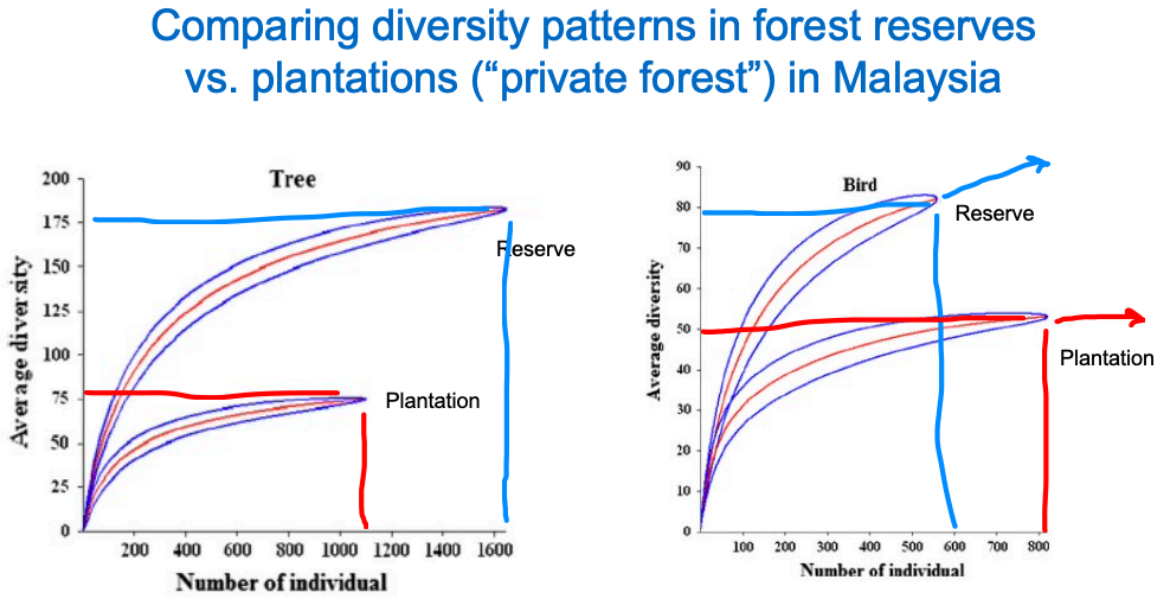
- Regional differences in species diversity are influenced by area and distance, which determine the balance between immigration and extinction rates

B. Species accumulation curves



- Species accumulation curves: Species richness plotted as a function of total number of individuals counted.

- These curves can help determine when most or all of the species in a community have been observed.
- In most real cases, the threshold where no new species are counted is never found because there are always new, rare species to add.

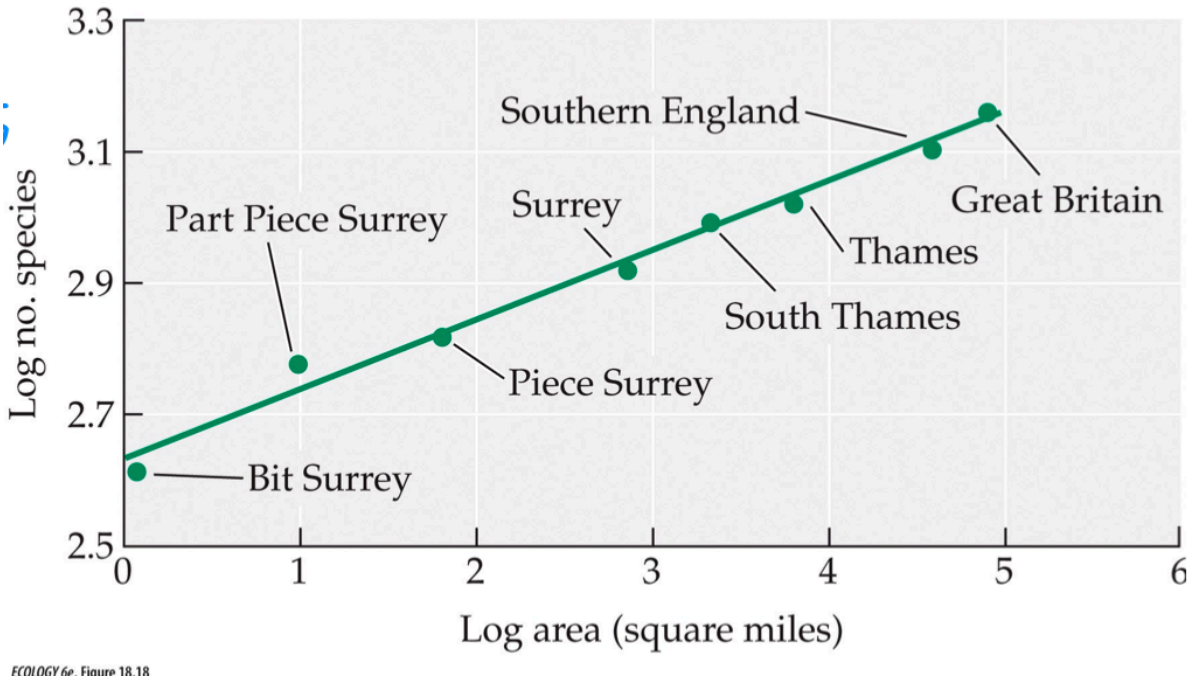


- there are likely more undetected bird species
- A fully detected one should be following the tree pattern

C. Species–area relationship:

- species richness increases with area sampled (assuming similar effort, e.g., similar number of individuals sampled in each area)

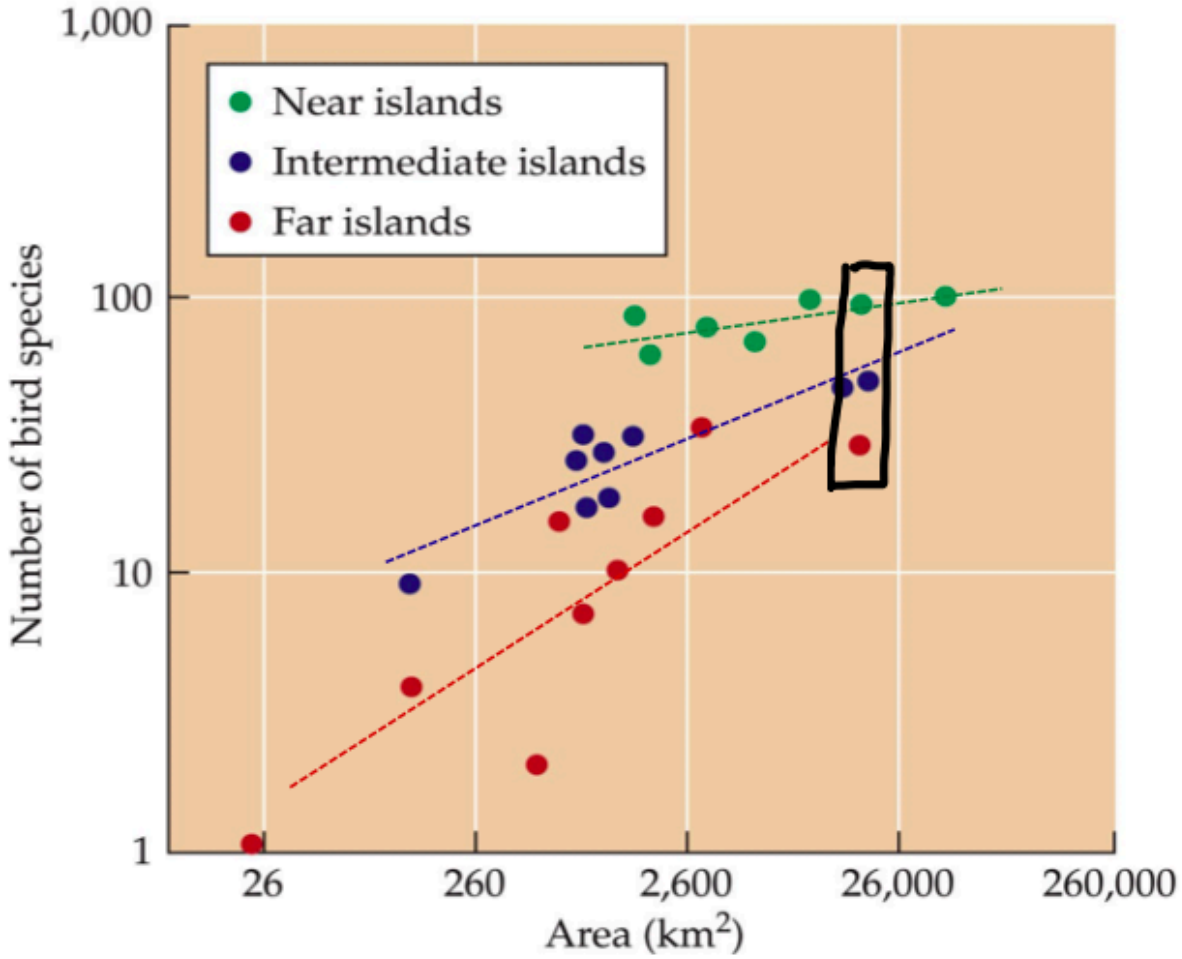
- The first species–area curve was made for plants in Britain in 1859.



ECOLOGY 6e, Figure 18.18

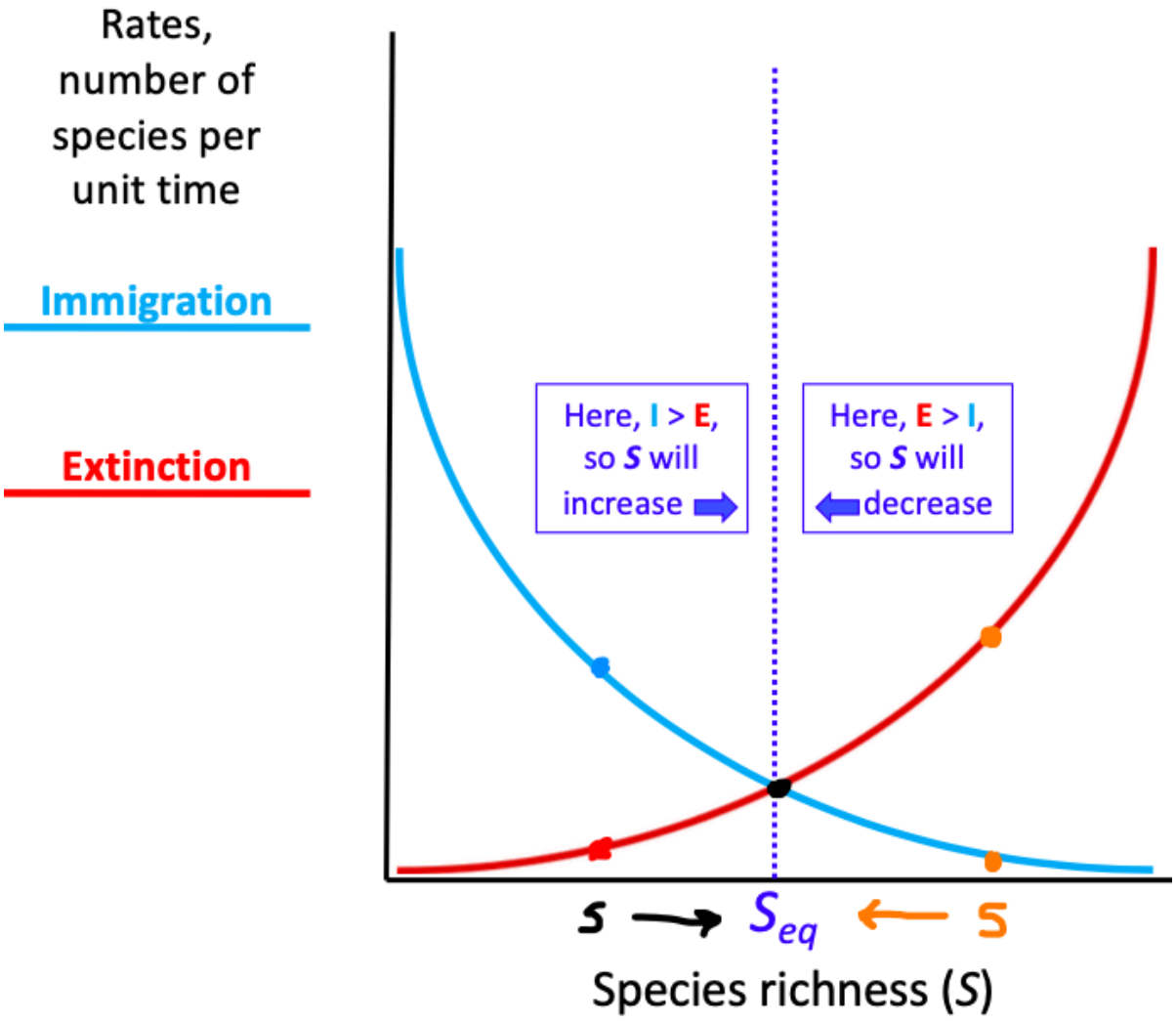
D. The Theory of Island Biogeography

- A method for asking how island area (size) and isolation (distance from the mainland) can act together to produce commonly observed species diversity patterns
- The theory assumes that all species are equivalent; there are no predator-prey relationships considered and no one species is a better competitor than any other



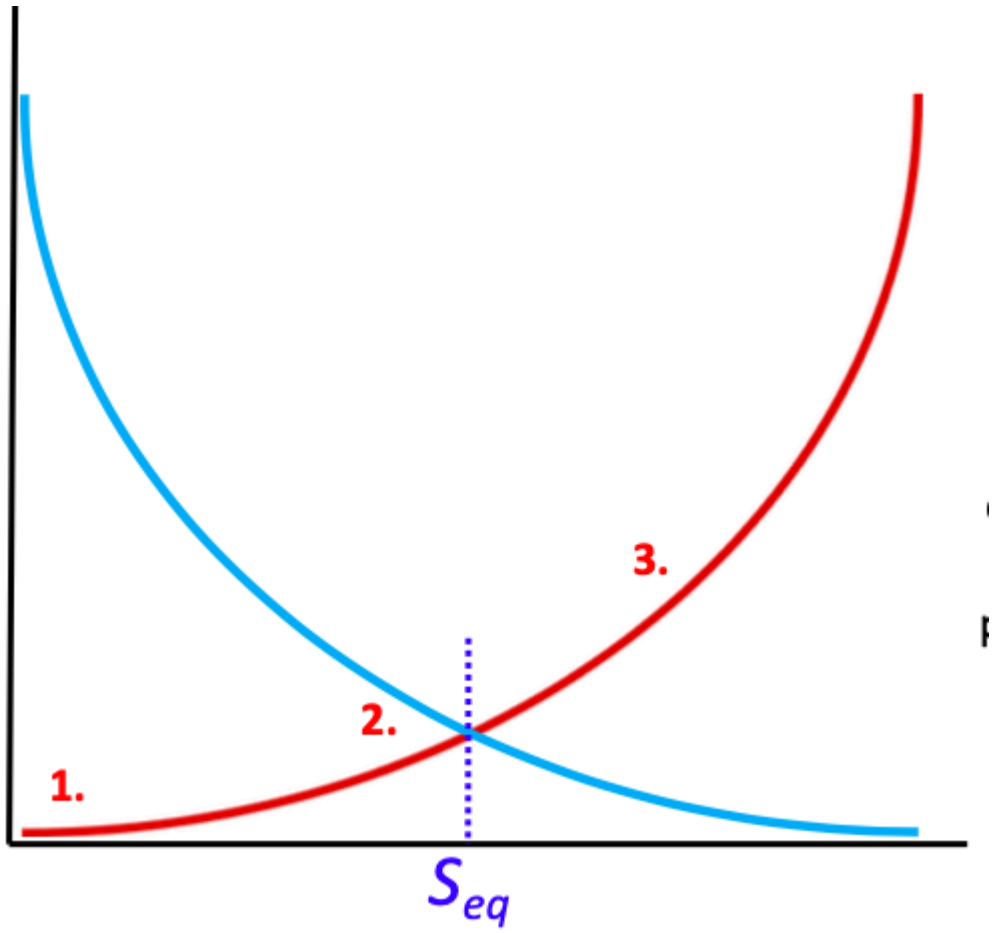
- population size, N, change through time: ΔN=B-D+I-E
- species richness, S, change through time: ΔS=speciation-extinction+immigration
    - Speciation-extinction: diversification
    - Extinction+immigration: key to the theory of island biogeography

① **Equilibrium model** of biogeography:

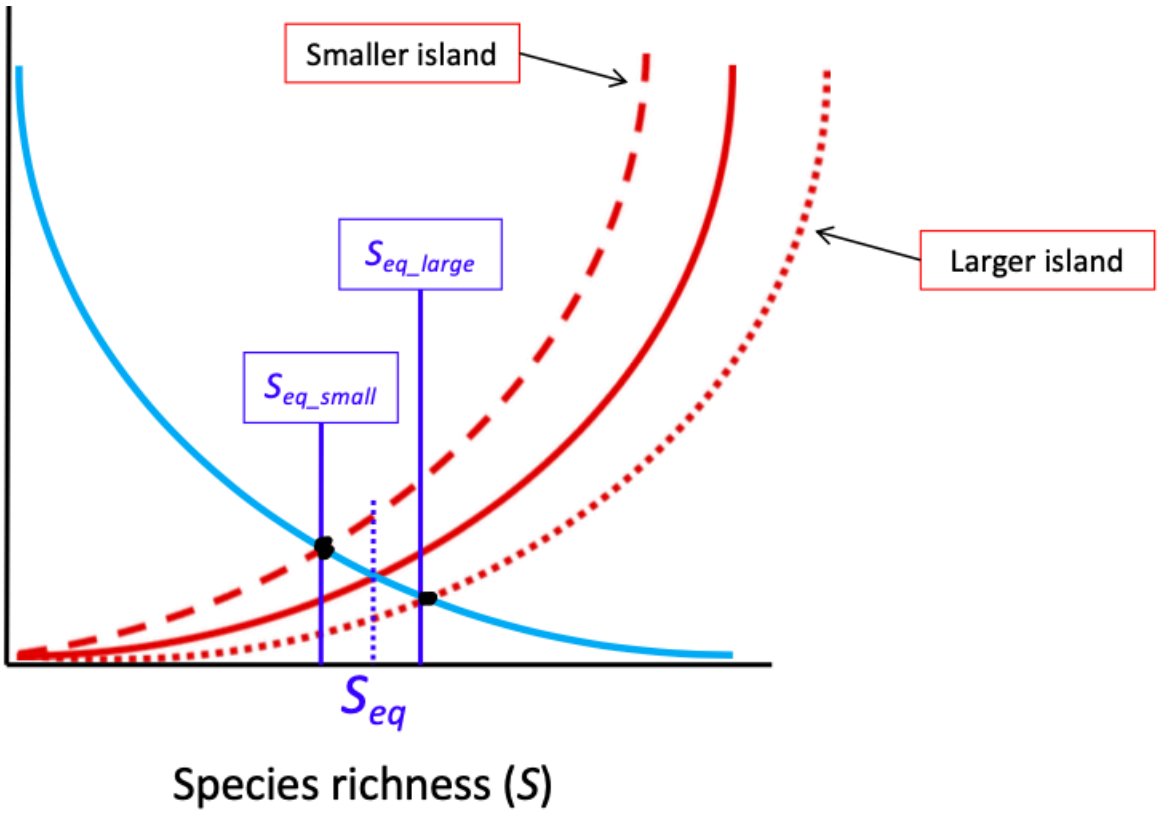


- At equilibrium, species richness (S) is stable due to balance between immigration and extinction.
- Disturbance reduces S below equilibrium:
    - Some species are lost
    - Open space/resources become available
    - Allows **new species to colonize** or **previously lost species to return**.
- If **extra species arrive** and push S **above equilibrium**:
    - Community is too crowded (resources limited).
    - **Higher chance of extinction** for either newcomers or existing species.

② **Why do the rates follow these curves (extinction)?**

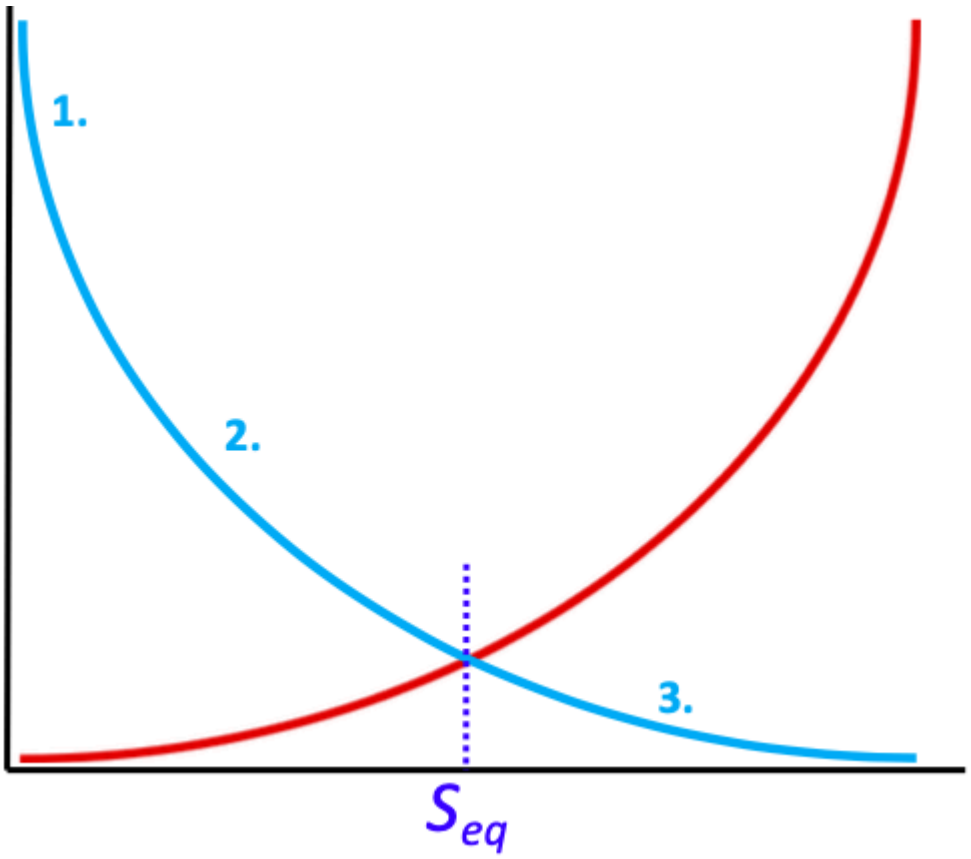


- 1. When S is small, life is easy, with relatively low competition and high population sizes. The extinction rate is low.
- 2. The larger S becomes, the more probable that resources will be used up efficiently. Species that have established will experience a reduction in their density, increasing the probability of extinction.
- 3. When S is very high, any disturbance, resource-shortage, or just random fluctuation in population size is increasingly likely to wipe out the small populations present; the extinction rate becomes very high.
- The extinction rate (or probability of extinction) is influenced mainly by area:



- Small islands:
    - Higher extinction rates (less space, fewer resources, smaller populations).
    - Result: Lower equilibrium species richness (S_eq).
- Large islands:
    - Lower extinction rates (more space, more resources, larger populations).
    - Result: Higher equilibrium species richness (S_eq).

③ **Why do the rates follow these curves (immigration)?**



1. When species richness (S) is low:

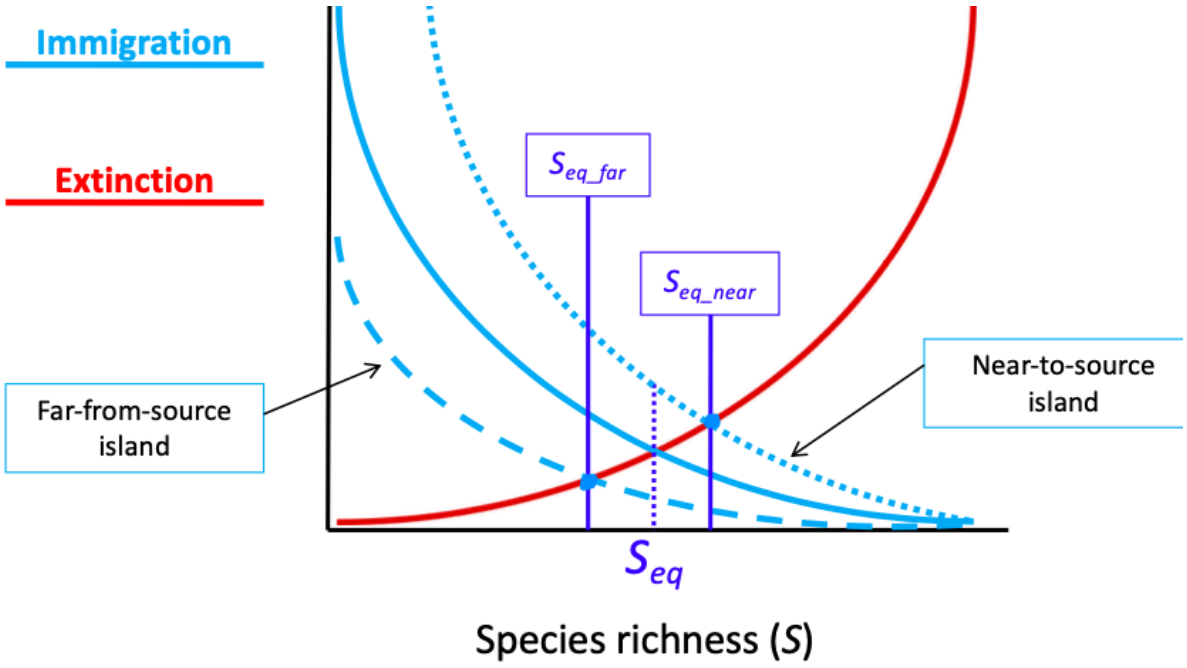
- Most new arrivals are likely to be new species.
- → High immigration rate of new species.

2. As S increases:

- Many species are already present.
- The chance that a new arrival is a new species goes down.
- → Immigration rate gradually decreases.

3. When S is high (near source species pool size):

- Few species remain that haven't already colonized.
- → “New” species run out; immigration rate approaches zero.



- The immigration rate is influenced mainly by isolation distance
- At **equilibrium (S_eq)**, **immigration and extinction continue**.
- → **Species composition is dynamic**, even when S stays constant.

14. Species richness and biogeography

A. Key concepts

- Patterns of species diversity and distribution vary at global, regional, and local spatial scales
- Global patterns of species diversity and composition are influenced by geographic area and isolation, evolutionary history, and global climate

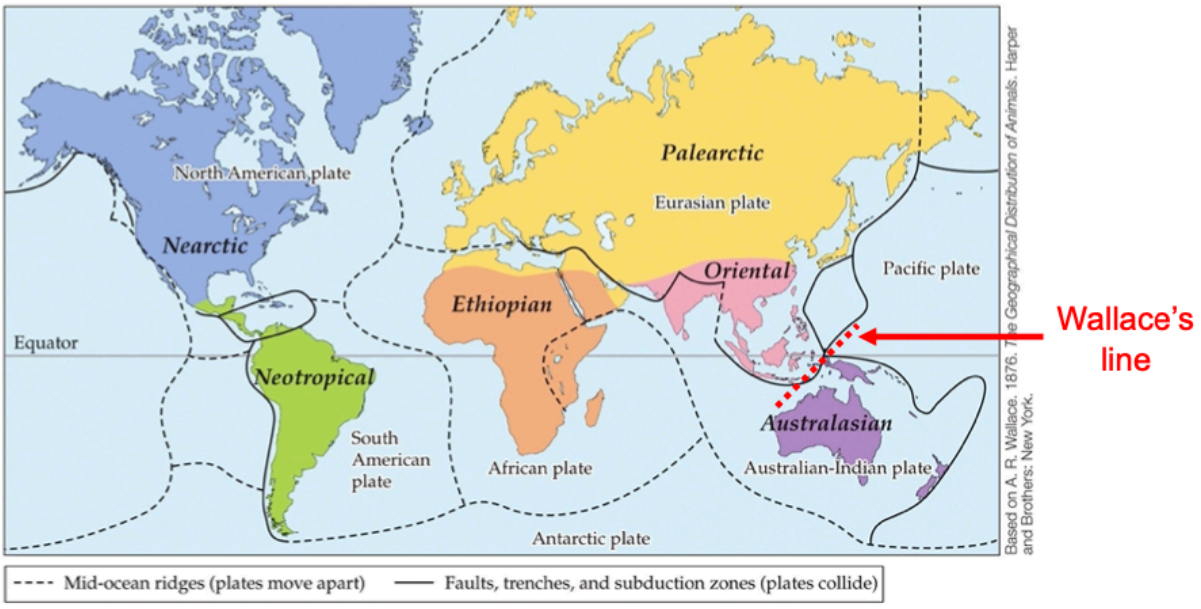
B. Latitude vs Diversity

- Forests around the world differ greatly in species richness and species composition.
    - Amazon rainforest: ~1,300 tree species → very high diversity.
    - Boreal forests (Alaska/Canada): Only ~2 dominant tree species → very low diversity.
- General patterns:

- **Species richness increases toward lower latitudes (near the equator).**
- Higher latitudes (toward poles) have fewer species.
- Regions at the same latitude can still differ in species richness and composition (e.g., Western North America vs. New Zealand).

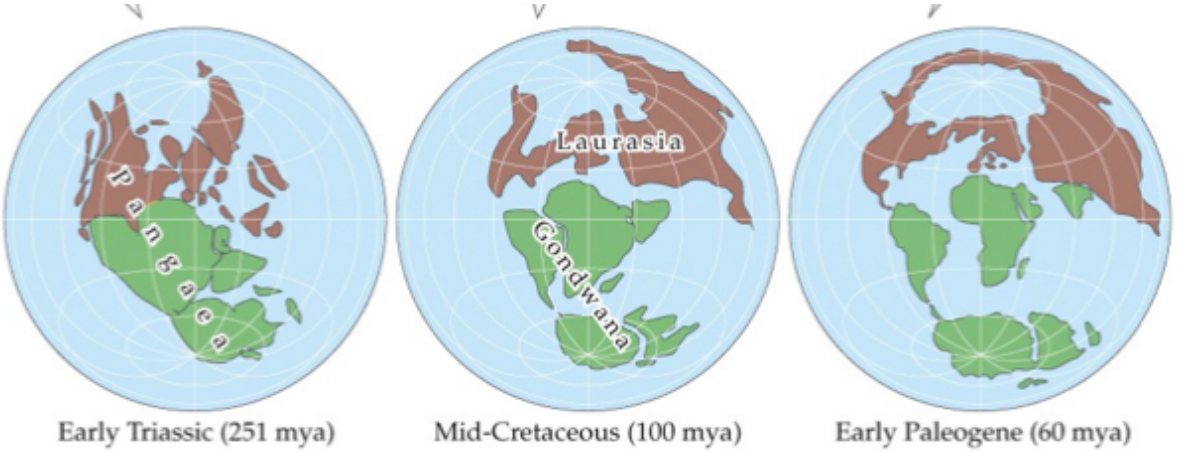
C. What controls these large scale, biogeographic patterns?

- **Alfred Russel Wallace (19th century)** is considered the father of biogeography.
- In the Malay Archipelago, Wallace observed:
  - Mammals in the Philippines were more similar to those in Africa than to those in nearby New Guinea.
  - The boundary separating these faunas is called Wallace's Line.
- In 1876, Wallace published The Geographical Distribution of Animals and identified two major global patterns:
  - The Earth's land can be grouped into six biogeographic regions.
  - Species richness increases toward the equator (latitudinal diversity gradient).



- The six biogeographic regions align roughly with Earth's major tectonic plates.
- Long-term isolation of regions led to unique flora and fauna.

- e.g., Neotropical, Ethiopian, and Australian regions have very distinctive species.
- North America (Nearctic) was part of Laurasia.
- South America (Neotropical) was part of Gondwana.
- The two were separate for most of history, only connecting via Central America about 3–6 million years ago.
- → Explains why their species differ greatly despite being close today.

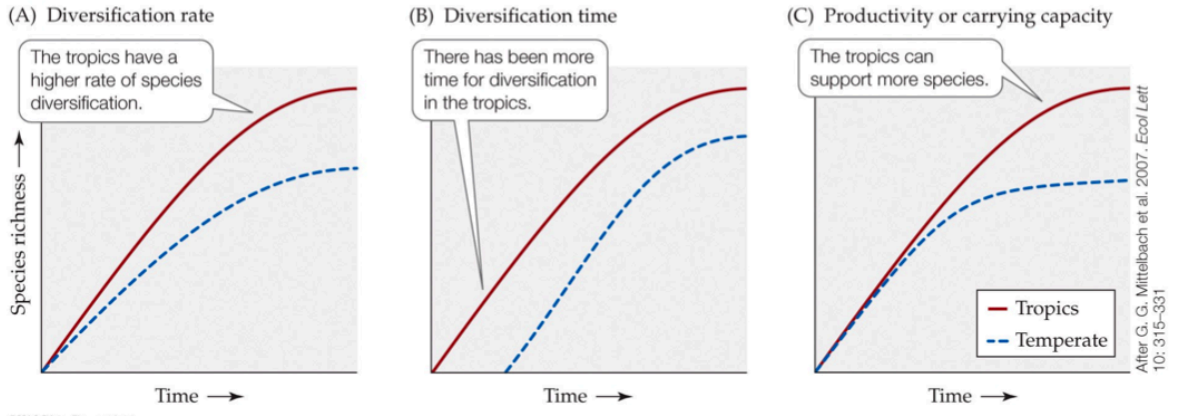


D. What drives the latitudinal diversity gradient?

- Latitudinal diversity gradient: more species at low latitudes than high latitudes.
- Species richness depends on the balance of:
  - Speciation rate
  - Extinction rate
  - Dispersal (movement among regions)
- If dispersal ≈ similar across regions, then:
  - Net diversification rate = speciation rate − extinction rate.
  - Determines whether species numbers increase or decrease over time.

Many hypotheses exist to explain **why net diversification is higher in the tropic**s, but no single consensus.

Most explanations fall into three main categories:



① Diversification rate hypothesis:

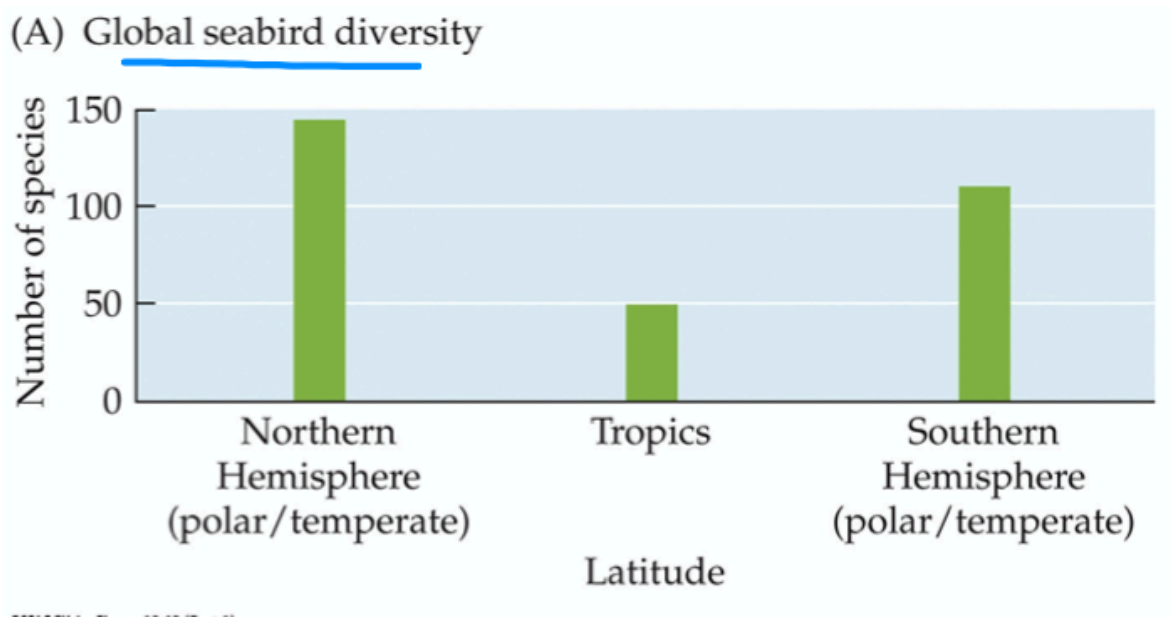
- The tropics have the most land area on Earth, and temperatures are very stable. Large, thermally stable areas may have higher diversification rates (diversification rate = speciation rate - extinction rate).

② Diversification time hypothesis

- The tropics are thought to have been more climatically stable over time, and species have had more time to evolve. Temperate and polar regions have undergone severe climatic changes such as glaciation , disrupting species diversification.

③ Productivity or carrying capacity hypothesis

- Because their productivity is higher, the tropics have a higher carrying capacity than temperate areas, so more species can coexist there. Productivity could also explain the reverse latitudinal pattern seen in sea birds; coastal oceans are most productive at higher latitudes.



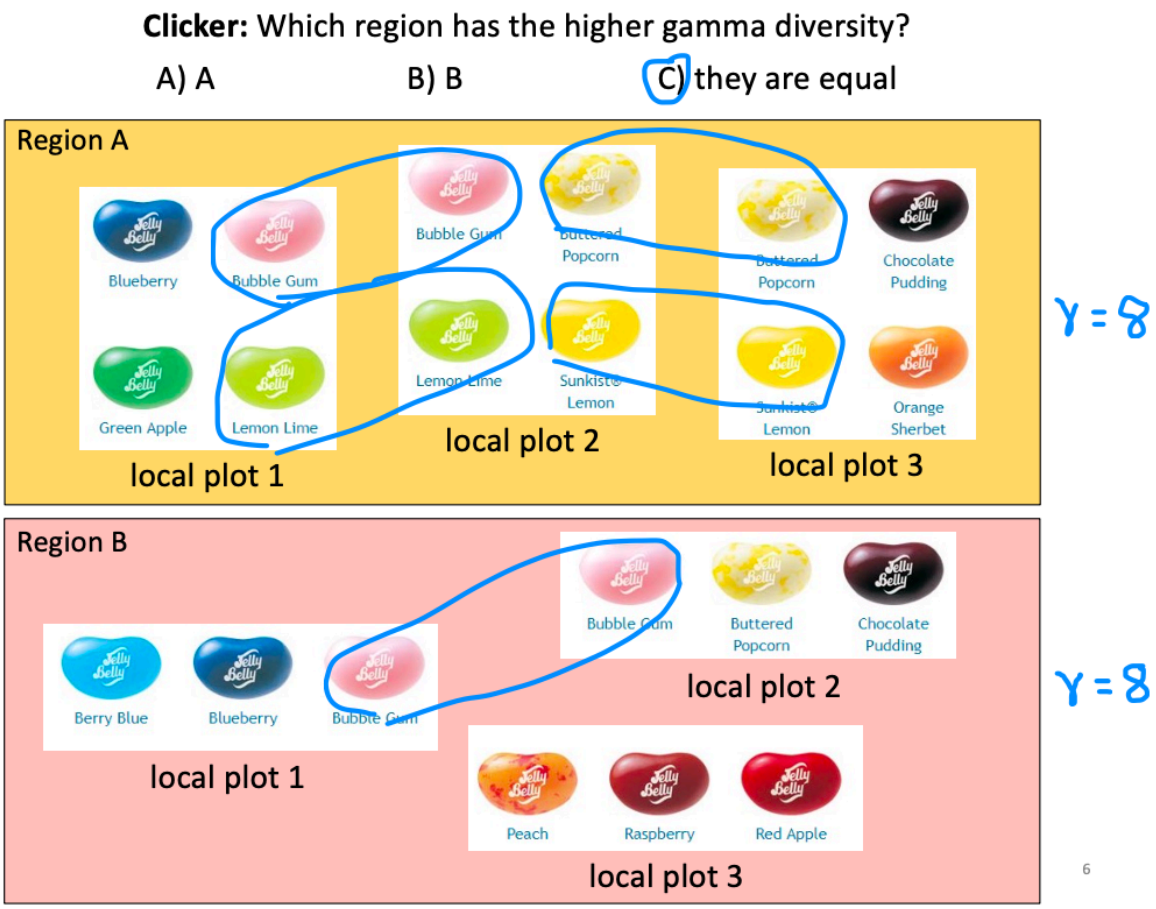
E. Interconnected spatial scales of species diversity (richness):

① Global scale:

- Global patterns of species diversity and composition are the result of variations in speciation, extinction, and dispersal at latitudinal and continental spatial scales and evolutionary time scales

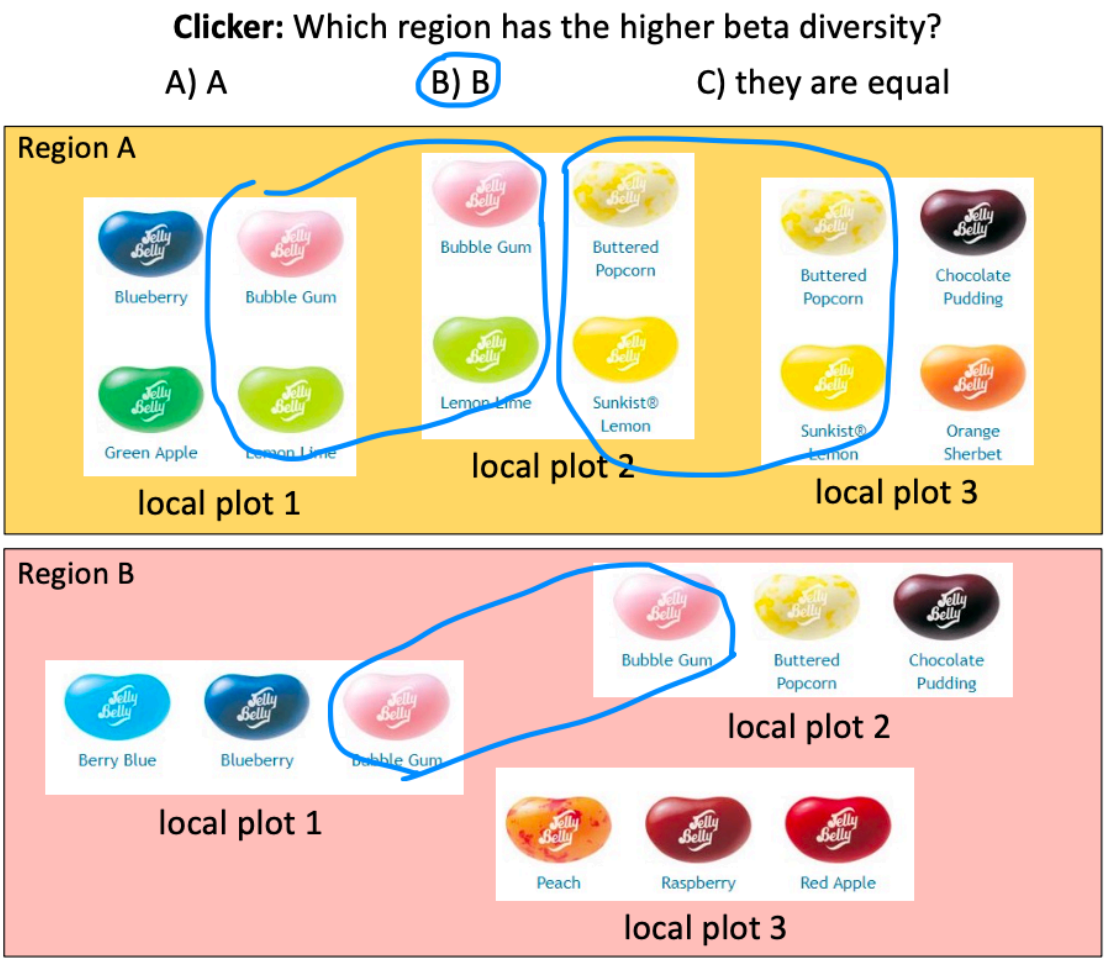
② Regional scale:

- Regional patterns of species diversity and composition (**gamma diversity**) are driven by dispersal across the landscape.



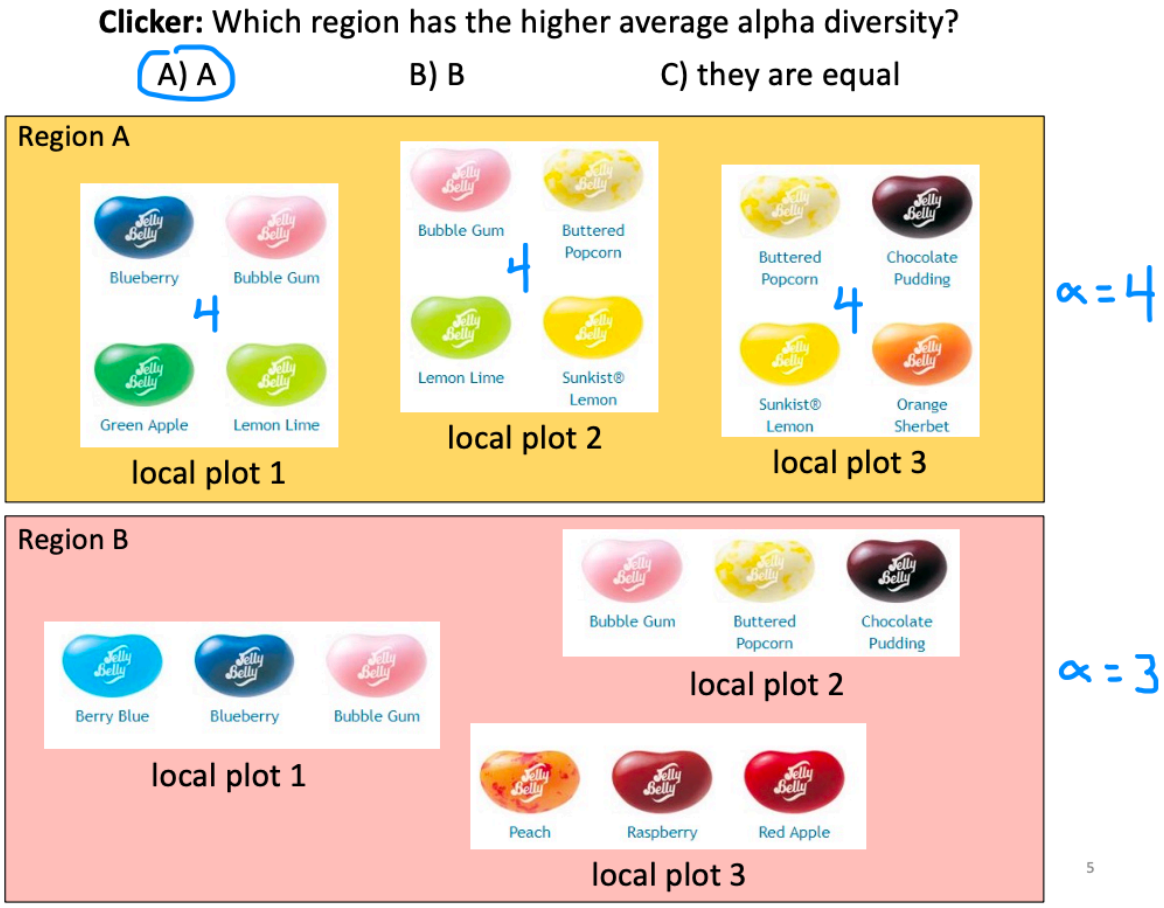
③ Landscape scale:

- Local and regional scales are connected by turnover, which is the difference in species diversity and composition (**beta diversity**) from one community type to another across the landscape.



④ Local scale:

- Local patterns of species diversity and composition (**alpha diversity**) are driven by physical conditions and species interactions.



$$\beta = \gamma / \alpha$$

beta = gamma / alpha

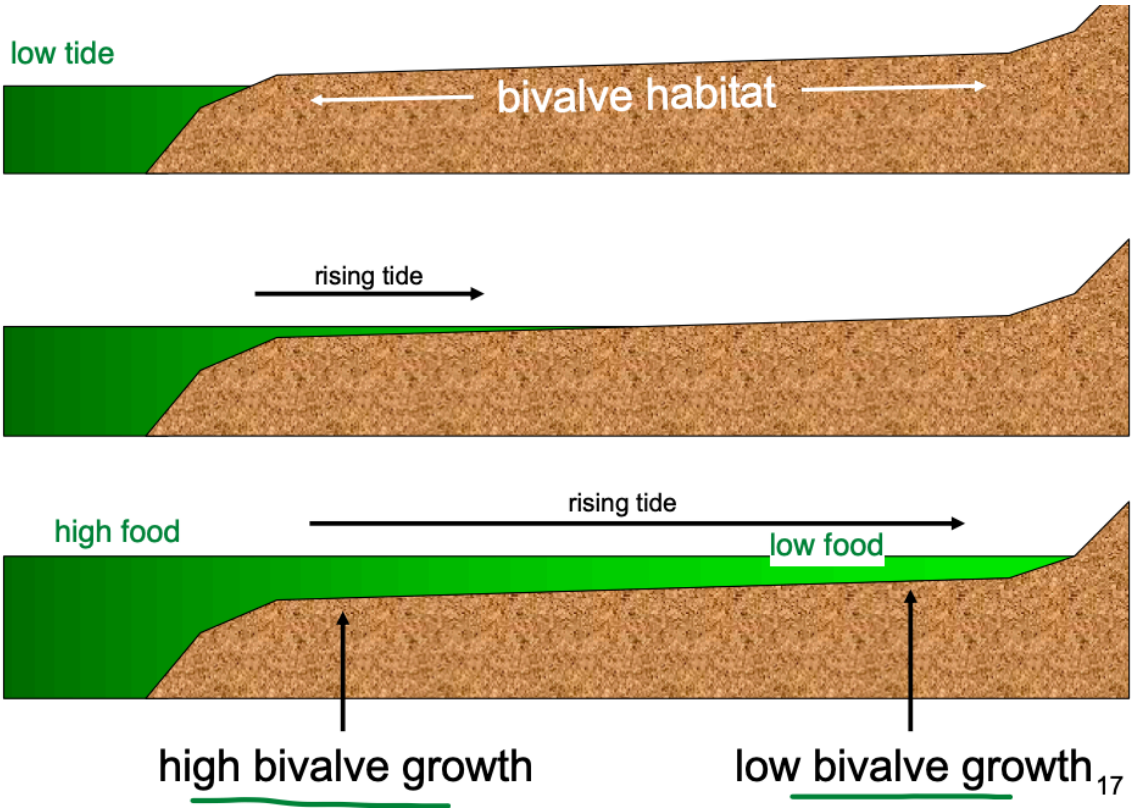
15. Competition

A. Key concepts

- Competition: a non-trophic interaction in which individuals of the same species (intraspecific) or different species (interspecific) are harmed by their shared use of a resource that limits their ability to grow, reproduce or survive.
    - Intraspecific competition – competition between individuals of the same species (which leads to density-dependent population growth and sets K)

- interspecific competition – competition between individuals of different species

- Competition can be direct or indirect, vary in its intensity, and occur between similar or dissimilar species
    - exploitation competition - reducing access to a resource by reducing the availability of that resource (an indirect effect)
        - Example: Scallops near the outer edge of the bay consume much of the planktonic food before it reaches the scallops in the inner bay. In this example, individuals can compete even when they are never in exactly the same place.



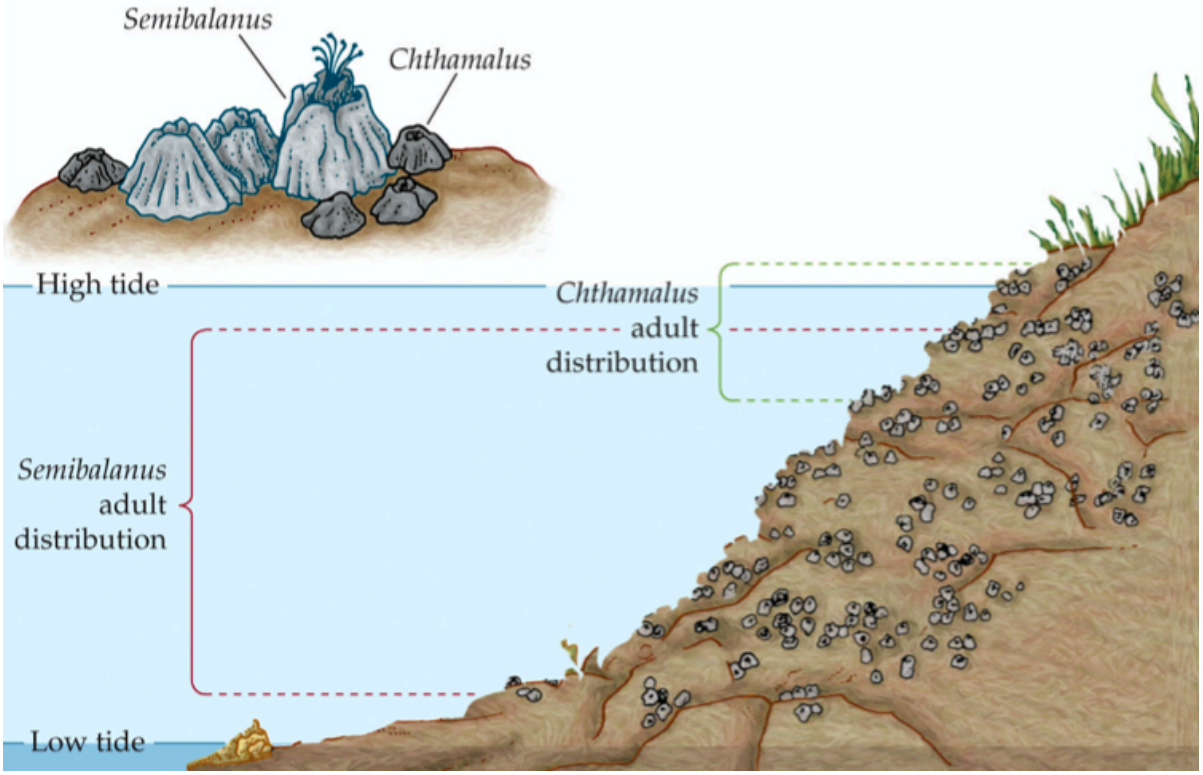
    - Interference competition - reducing access to a resource* by directly interfering with a competitor (a direct effect)
    - Resources: components of the environment (e.g., food, water, light, space) that are required by species and can be depleted by them

- Competing species are more likely to coexist when they use resources in different ways
- The outcome of competition can be altered by the physical environment, disturbance, and predation

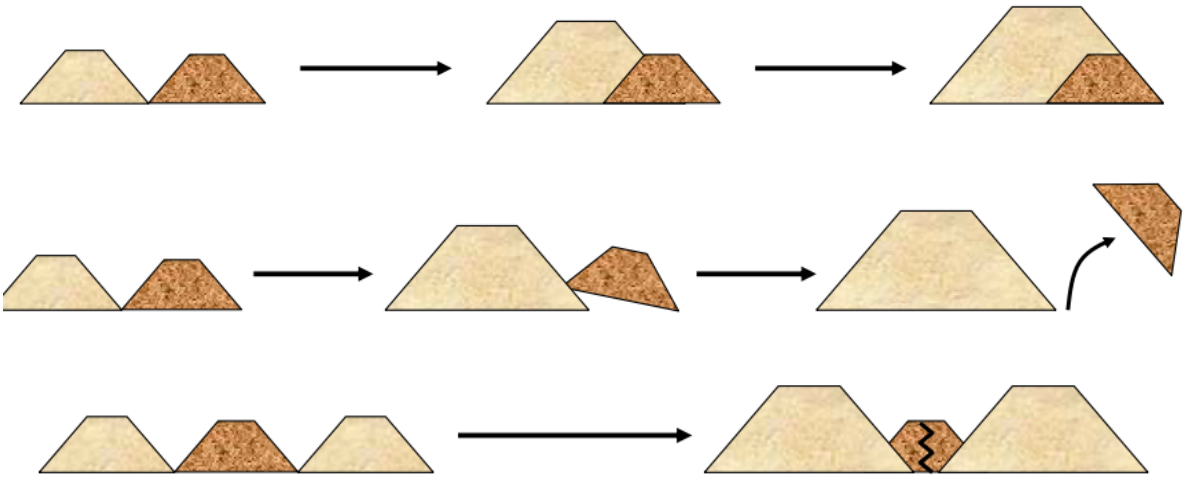
B. Asymmetric interspecific competition

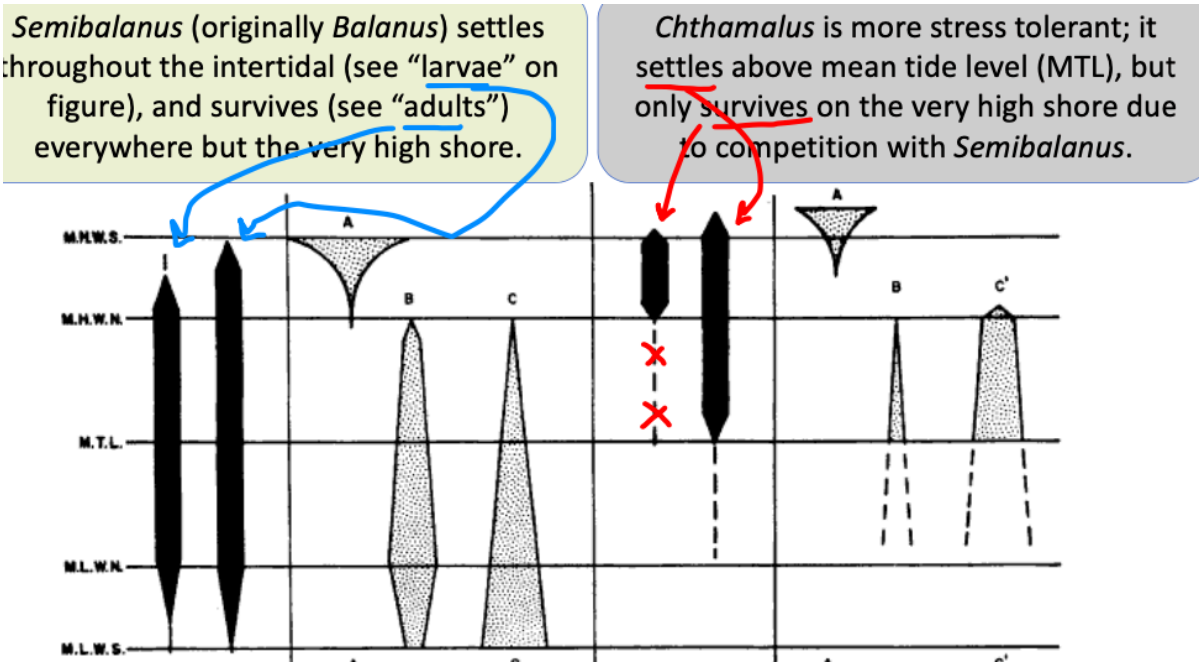
- Competition reduces the resources available for both species.
- But effects of competition are often unequal, or asymmetrical; one species is harmed more than the other
- In some cases, notably when two species share the same limiting resource, one species can drive another to extinction.
- **competitive exclusion principle**: two species that use a limiting resource in the same way cannot coexist indefinitely.

C. Case study: competitive exclusion in two species that compete for space along a gradient of intertidal height



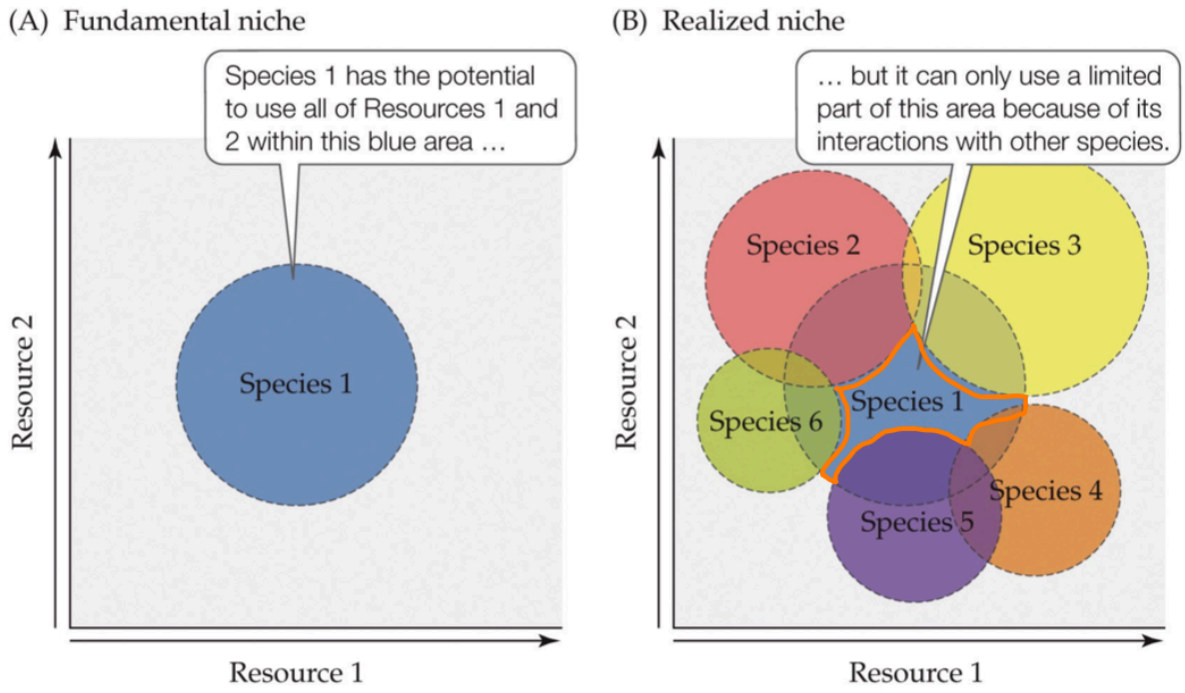
- Interference ways:





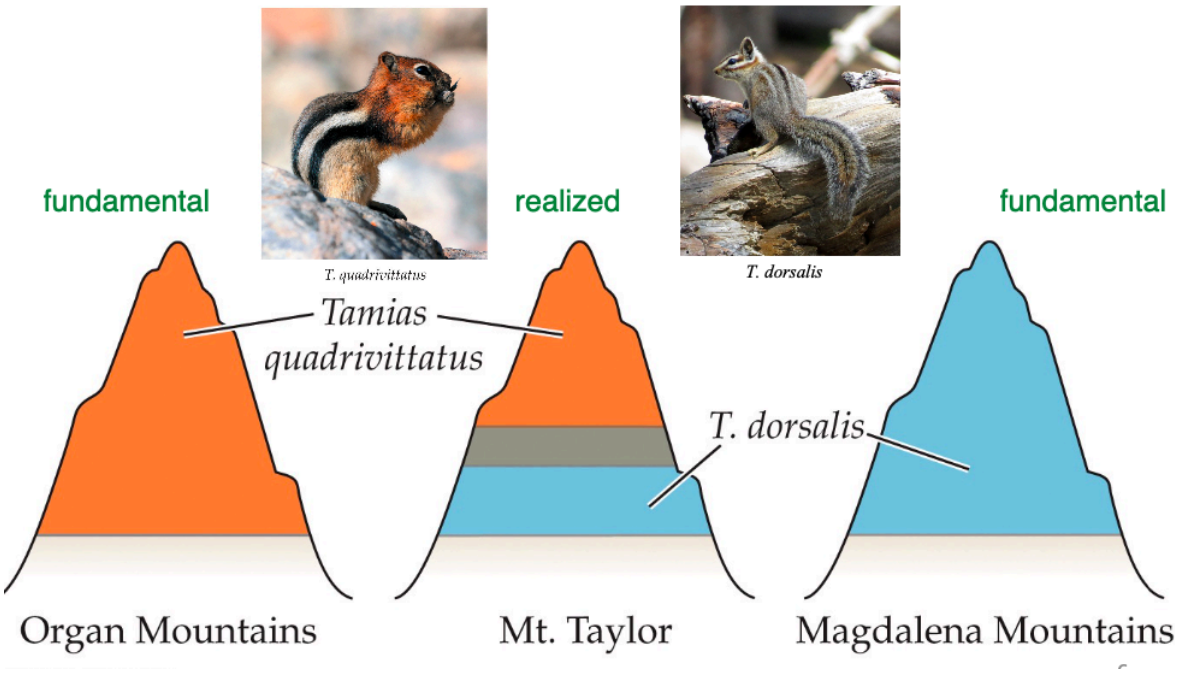
D. Niches

- Competition plays an important role in limiting the distribution of species as well as their abundance.
- Fundamental niche: the full set of resources, plus other biotic and abiotic requirements of a species, that allow it to survive, grow, and reproduce.
- Realized niche: the restricted set of resources or conditions a species is limited to, due to species interactions.



16. Competition (continued)

A. Niches Continued



Implications:

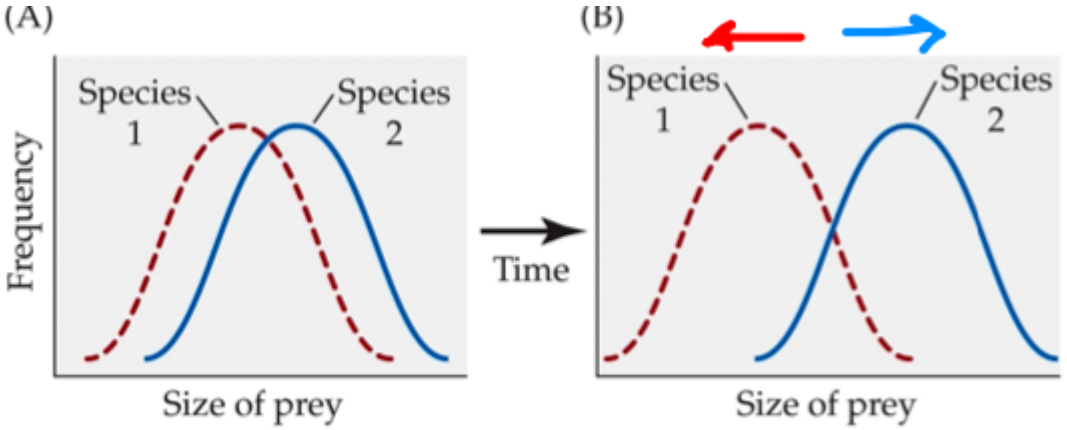
1) Both species can live at all elevations, but

2) neither species wins in competition at all elevations. Therefore,

3) something about elevation tips the competitive balance.

B. Resource partitioning: species use a limited resource in different ways

- a way to avoid competitive exclusion while still living in the same area
    - competitive exclusion: the predicted outcome of two species using the same limiting resource in the same way
- Eg. green and red light absorbing cyanobacteria competing for white light?
- Eg. Galapagos finches. Small beaks are specialized for small seeds; large beaks are specialized for large seeds.
- Character displacement: the shift in phenotypes through time in response to competition
    - Character displacement reduces the strength of interspecific competition



- Differences in the degree of overlap in resource use is one factor that determines how strong or weak competition is, and how asymmetrical it is.
    - Other factors include how good you are at finding, acquiring, and defending a resource, and how fully you can deplete it

C. Lotka-Volterra competition models

To predict coexistence and competitive exclusion based on knowledge of carrying capacity and the strength of interspecific competition.

- ultimate results of competition are changes in the population growth rates and population size of competing species
- Species in isolation:

density dependent term

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)$$

N_2

α

- Species in competition:

population size of

competition

$$\frac{dN_1}{dt} = r_1 N_1\left(1 - \frac{(N_1 + \alpha N_2)}{K_1}\right)$$

- competition coefficient α = per capita negative competitive effect of species 2 on species 1
- α = # of individuals of species 1 that are lost for every extra individual of species 2
    - if alpha<1,inter <intra; if alpha >1,inter >intra
- Species 2:

$$\frac{dN_2}{dt} = r_2 N_2 \left(\frac{K_2 - N_2 - \beta N_1}{K_2}\right)$$

- competition coefficient β = per capita negative competitive effect of species 1 on species 2
- β = # of individuals of species 2 that are lost for every extra individual of species 1

D. zero population growth isocline

To know when the population size is stable(rate=0):

$$0 = \frac{dN_1}{dt} = r_1 N_1 \left(\frac{K_1 - N_1 - \alpha N_2}{K_1} \right)$$

rstanding
n size is
know the
e, what do

$$0 = r_1 N_1 \left(\frac{K_1 - N_1 - \alpha N_2}{K_1} \right)$$

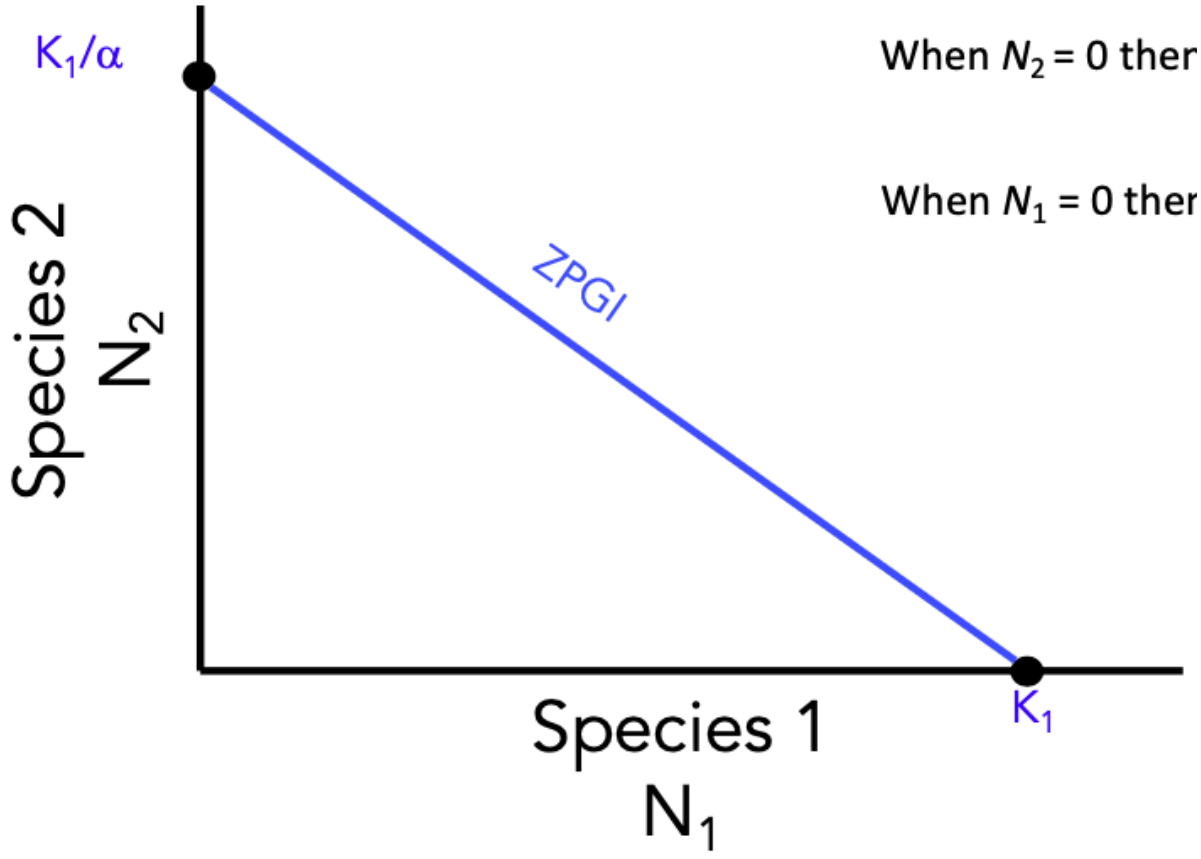
$$0 = \left(\frac{K_1 - N_1 - \alpha N_2}{K_1} \right)$$

$$0 = K_1 - N_1 - \alpha N_2$$

$$N_1 = K_1 - \alpha N_2$$

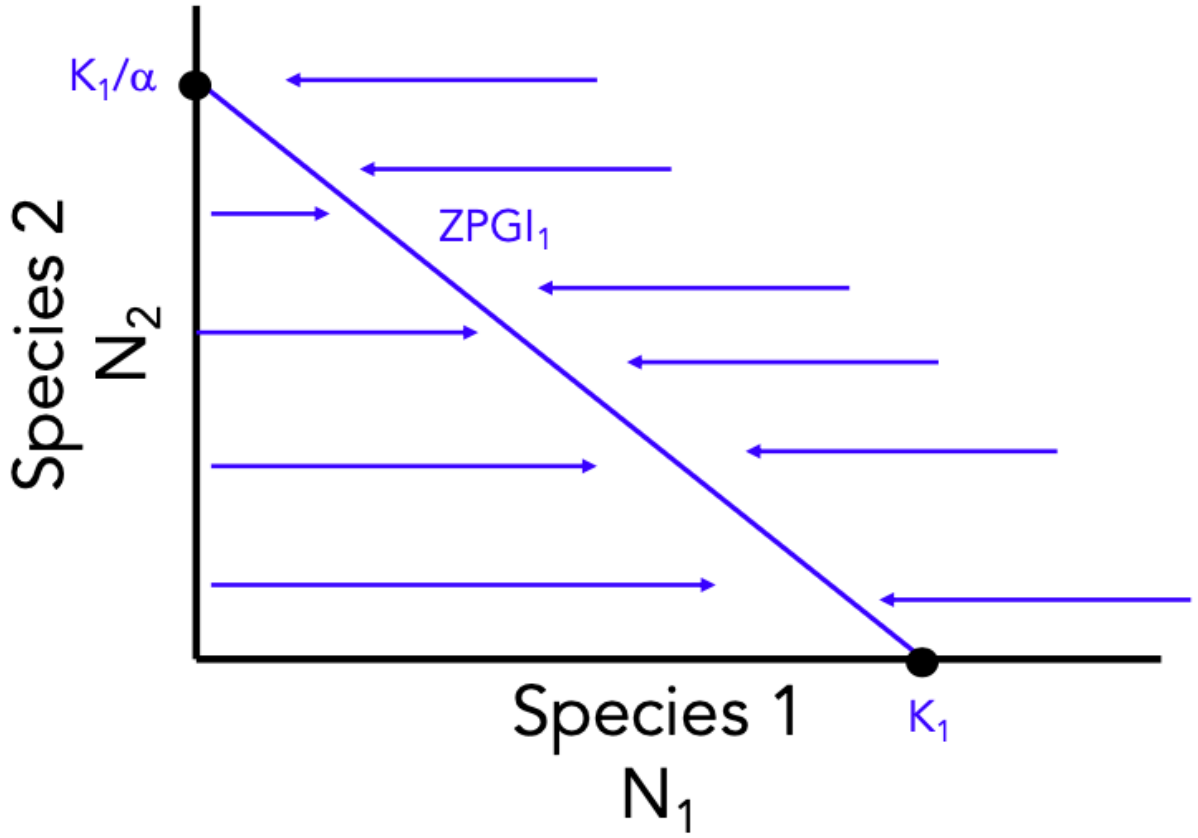
equilibrium population density

- When N2 = 0 then N1 = K1
- When N1 =0 then N2 =K1/α



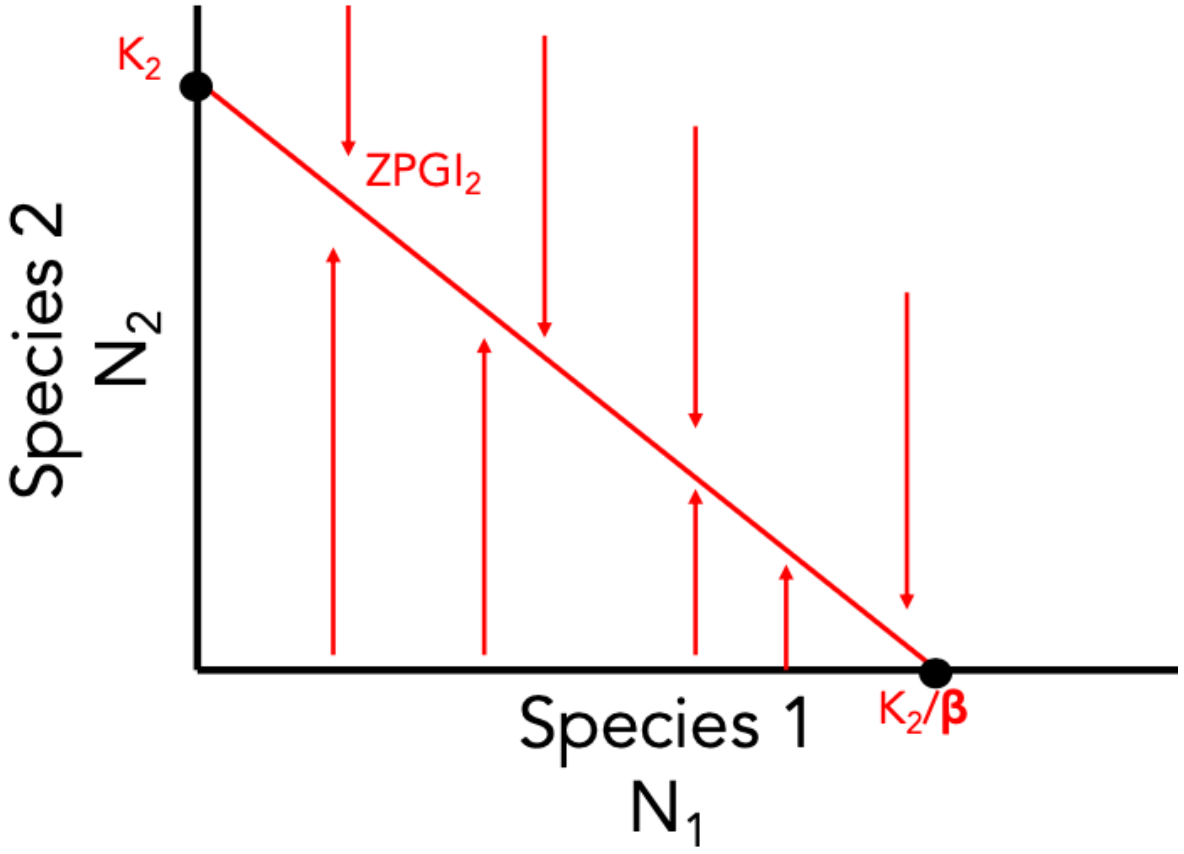
- **Analyzing Lotka-Volterra growth models:**

1. Examine the zero population growth isocline (ZPGI) for species 1
    a. N1 = K1 −αN2



2. Examine ZPGI for species 2

    a. N2 = K2 – βN1

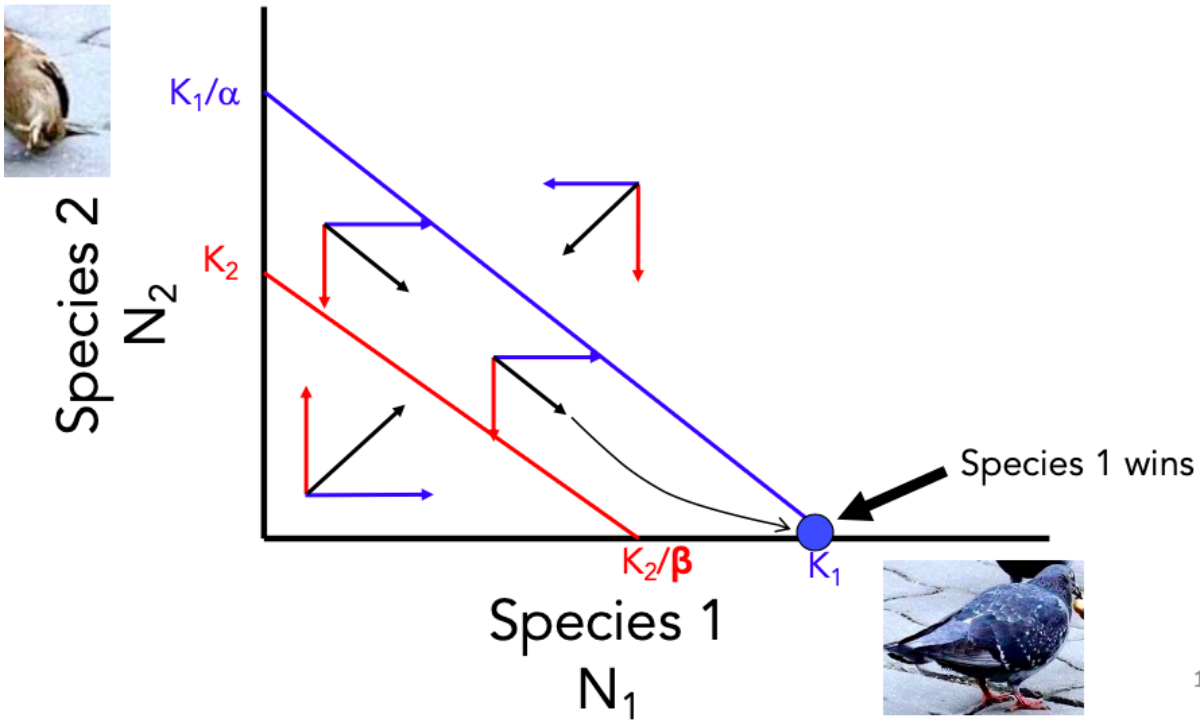


17. Competition (concluded)

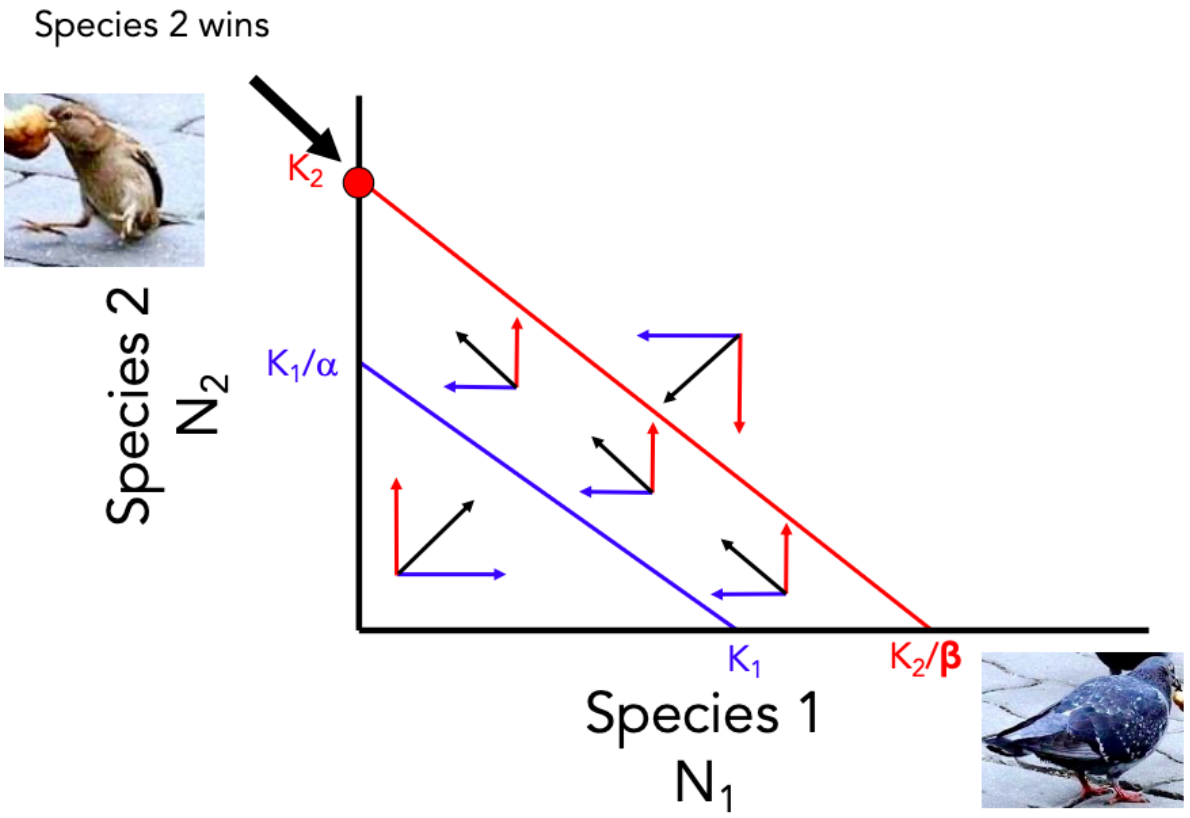
A. zero population growth isocline continued

3. Join isoclines on the same graph

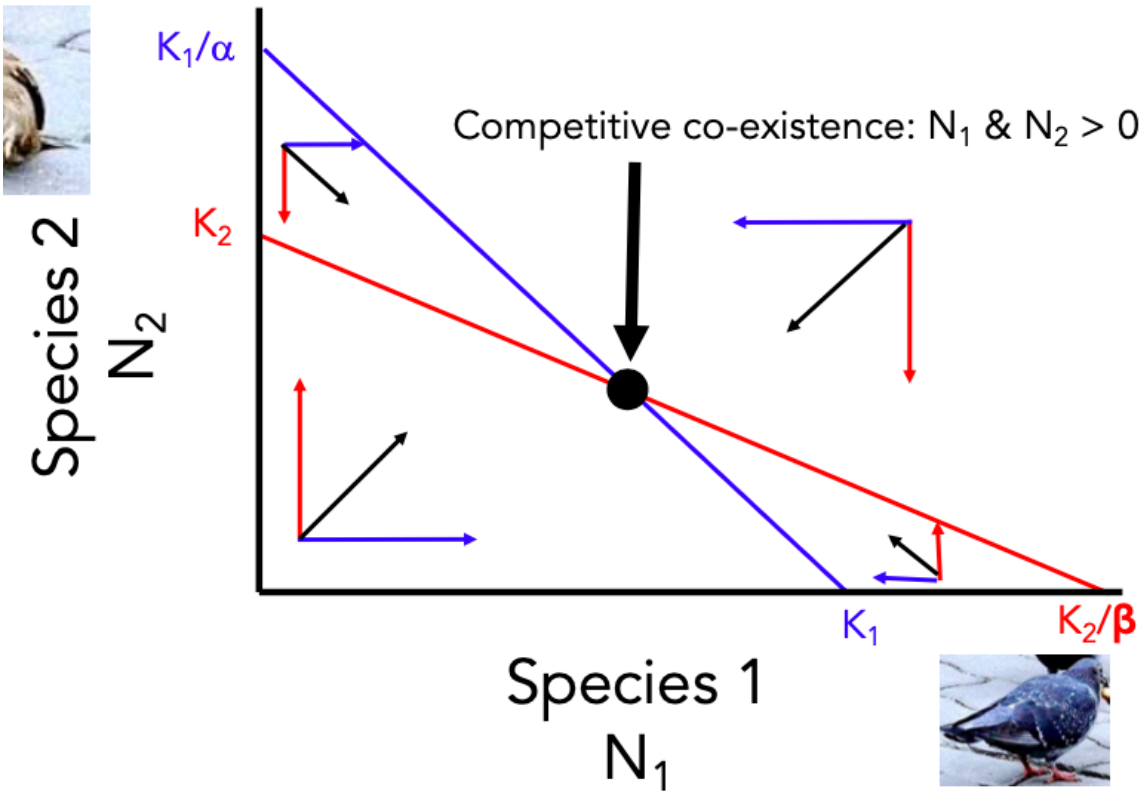
    a. If species 1 wins:



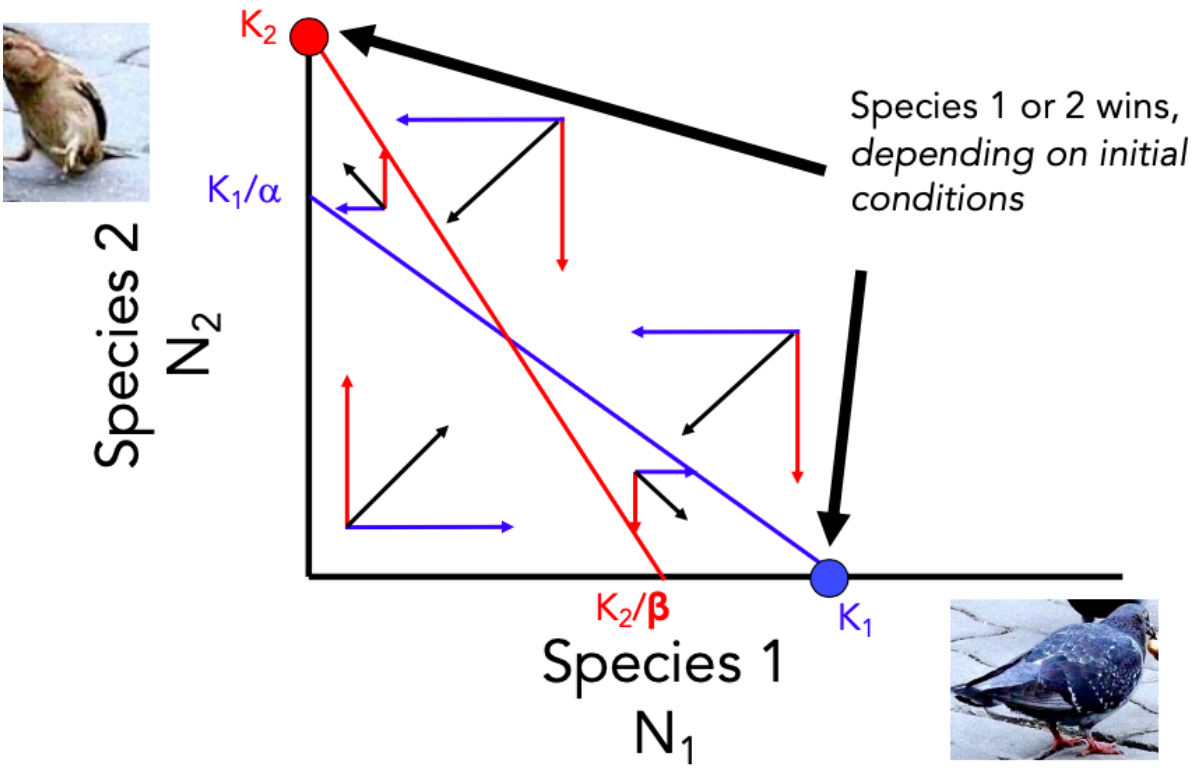
b. If species 2 wins:

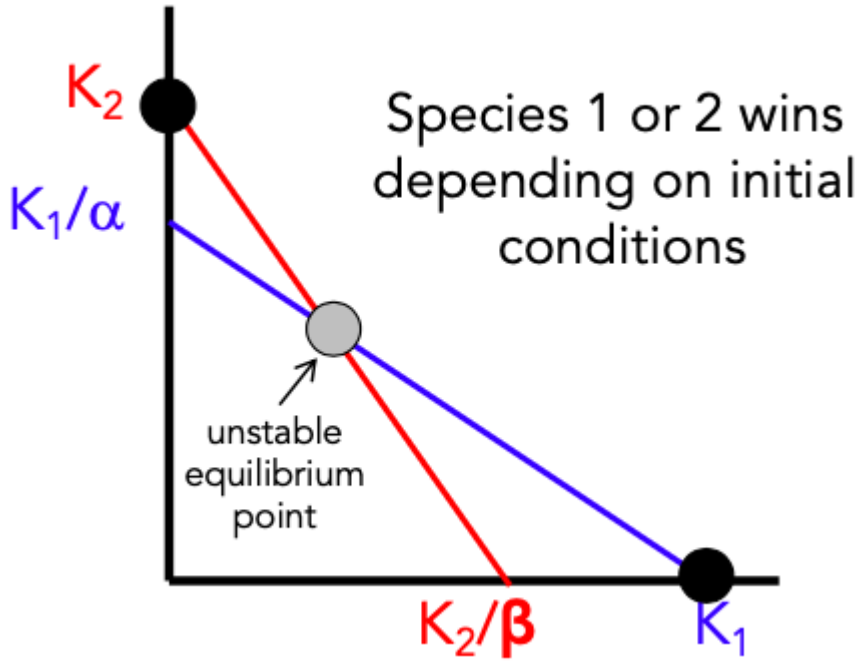


c. If Stable co-existence:



d. If unstable co-existence:

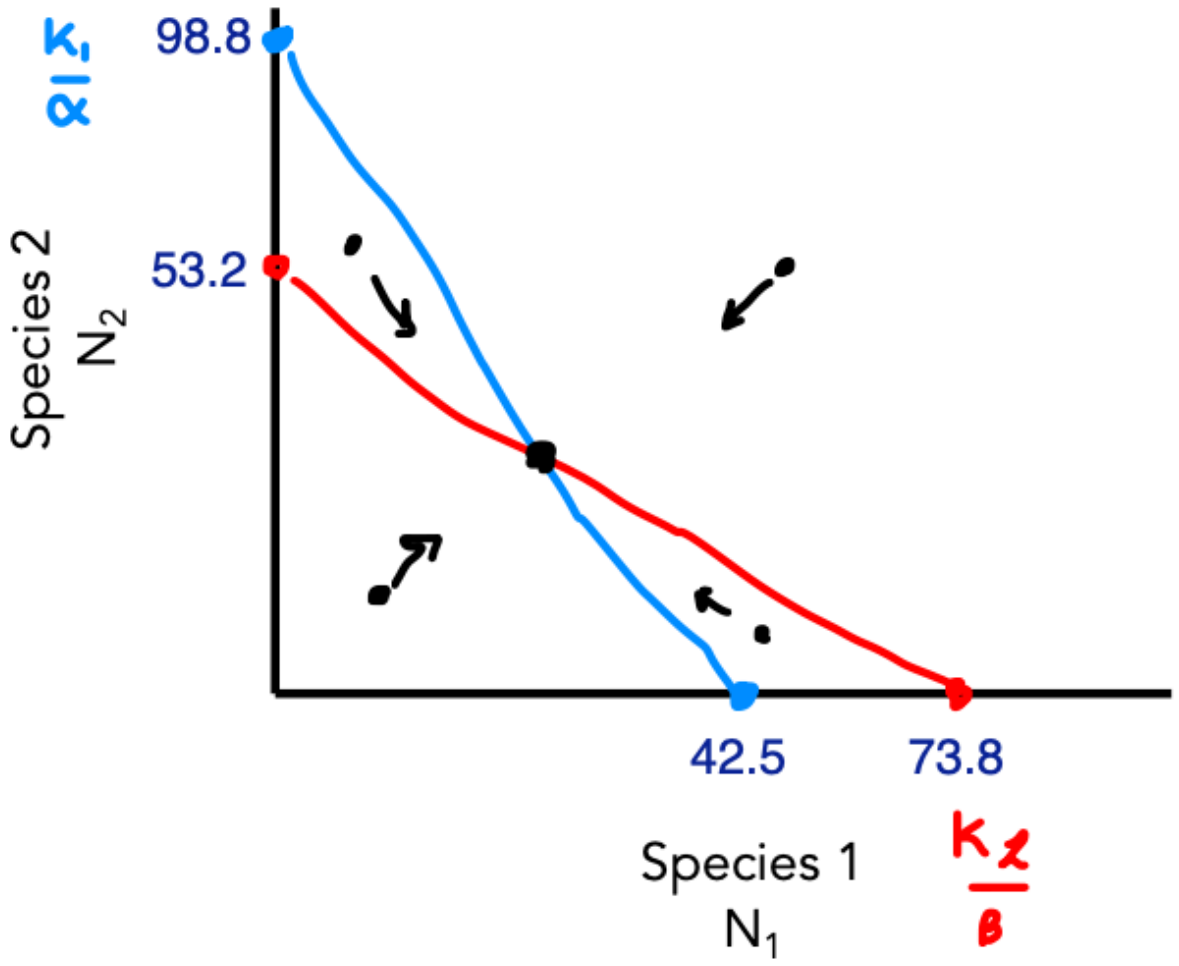




B. Case Study:

Ecologists were trying to determine if a newly introduced species of mosquito (Sp. 2) would be competitively excluded by a species that was already present (Sp. 1), with collected data:

$\alpha = 0.43$, $\beta = 0.72$, K1 = 42.5, K2 = 53.2 ——> coexist



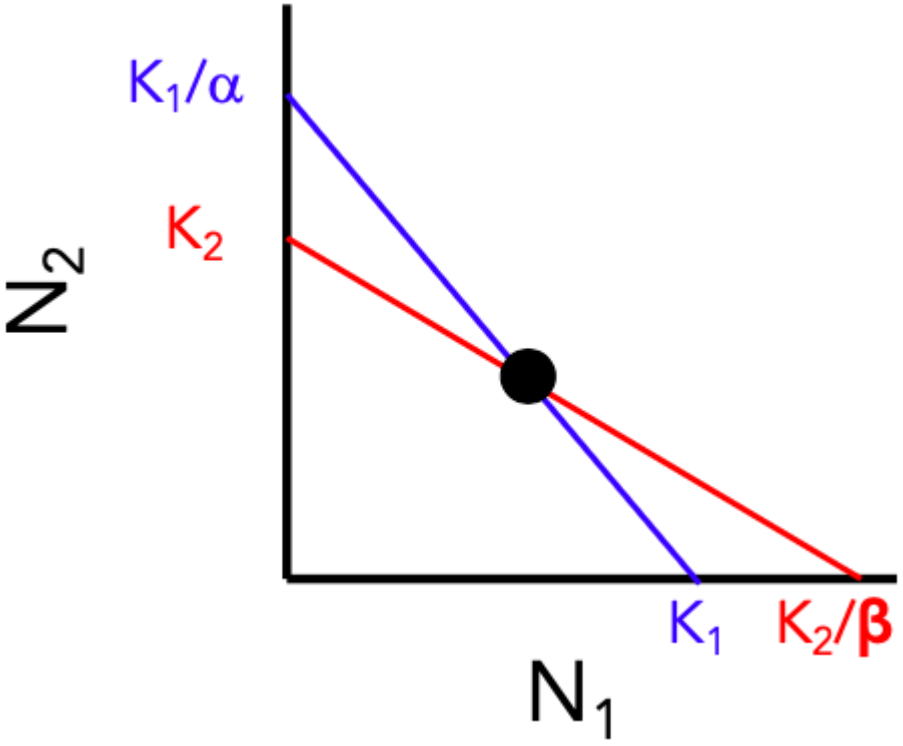
C. Competitive exclusion

- When the isoclines do not cross, competitive exclusion always results
- Which species wins depends on which isocline is above and to the right of the other

- OR
  - The isoclines do cross, but each species is more limited by the other than it is by itself (α and $\beta > 1$)
  - The winner depends on initial conditions
- Note that there are two ways to "win" and competitively exclude the other species
  - Be a stronger competitor
  - Have a higher carrying capacity

Stable coexistence:

- Zero population growth isoclines intersect and
- One or both species must inhibit itself more than it is inhibited by other species
    - Intraspecific competition > interspecific competition



- Possible when $\alpha < K1/K2 < 1/\beta$

What if α and β are below 1, but very near to 1?

- Species 1 has nearly the same effect on species 2 as species 2 does on itself, and vice versa
- Species use resources very similarly
- Carrying capacities must be very similar for coexistence

What if α and β are << 1?

- Species 1 has a much weaker effect on species 2 than species 2 does on itself, and vice versa
- Species use resources very differently

- Carrying capacities can be very different and coexistence may still be possible

Strong, asymmetrical competition does not guarantee competitive exclusion

- $\alpha < K1/K2 < 1/\beta$
- Can both species coexist if $b > 1$? Yes, as long as
    - a is sufficiently low and
    - K2 is enough higher than K1

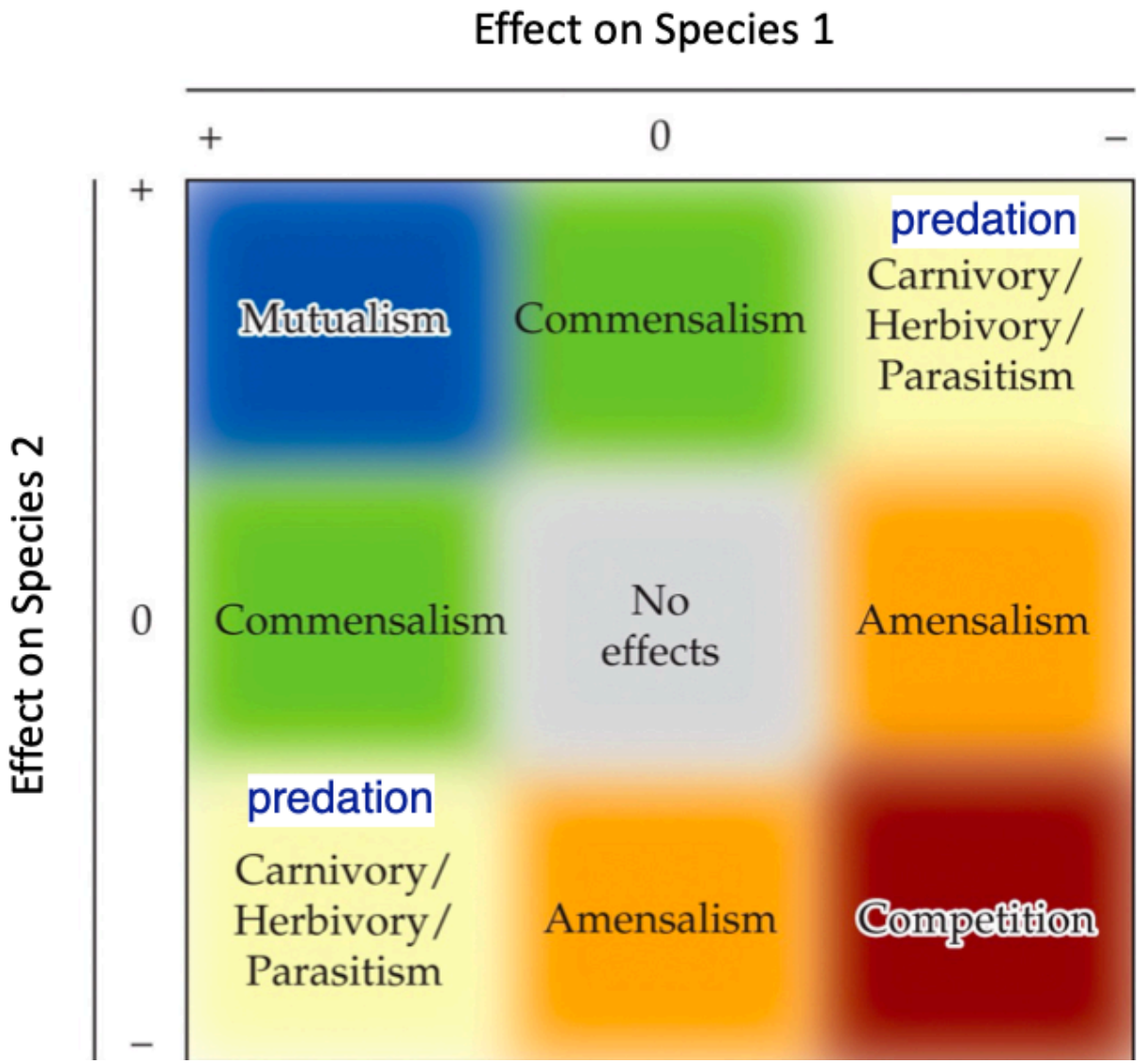
D. Recap: what to know about competition

1. Competition reduces the growth, reproduction, and/or survival of both competitors
2. When two species use the same limiting resource in the same way, ecological theory predicts that they cannot coexist indefinitely due to eventual competitive exclusion
3. In order to survive, species may have to adjust where they live or how they live, creating ecological pattern
4. Character displacement reduces a and b
5. Coexistence can occur given the appropriate balance between the strength of competition between species and differences in their carrying capacity
6. (...and coming up), competitive exclusion can be thwarted by forces that differentially impact the competitive dominant

18. Predation

A. Key Concepts

- Interspecific Interactions: Interactions between species that can affect growth, survival, and fitness



- Most carnivores have broad diets, whereas a majority of herbivores have relatively narrow diets
- Predation results in a wide range of capture and avoidance mechanisms
- Predation (including herbivory) can affect prey distribution and abundance, in some cases causing a shift from one community type to another

B. Predation

- Predation: a trophic interaction in which an individual of one species (the predator) consumes some or all of an individual of another species (the prey).
    - Carnivory—both predator (and some plants) and prey are animals.

- Herbivory—predator is an animal, prey are plants or algae.
- Parasitism—predator (a parasite) lives symbiotically on or in the prey (its host) and consumes certain tissues; may not kill the host. Some parasites (pathogens) cause disease.

C. Optimal foraging theory

Carnivores vs Herbivores:

- Animal prey are generally more nutritious than plant prey (nitrogen content higher)
- Plant prey are generally more abundant and easier to catch
- Dietary preferences will depend on:
    - Encounter rate — if low (typical of carnivores), predators should be generalists.
    - Handling time — if prey are easy to find but take a long time to eat (typical of herbivores consuming immobile but less nutritious plants), then predators (herbivores) should be specialists on particular plants or plant parts.
    - E.g. herbivorous fly diets (most eat 1-2 species of plants) and carnivorous sea star diets (e.g., Pisaster) that eat dozens of species, depending on what's available

D. Predator vs Prey

Steps to getting a good meal (Predator):

1. locate prey —--- visual/chemical/acoustic/electrical
2. catch prey —--- speed/dexterity
3. consume prey —---- handling time
4. digest prey —---- digestive physiology

Steps to not becoming a meal (Prey):

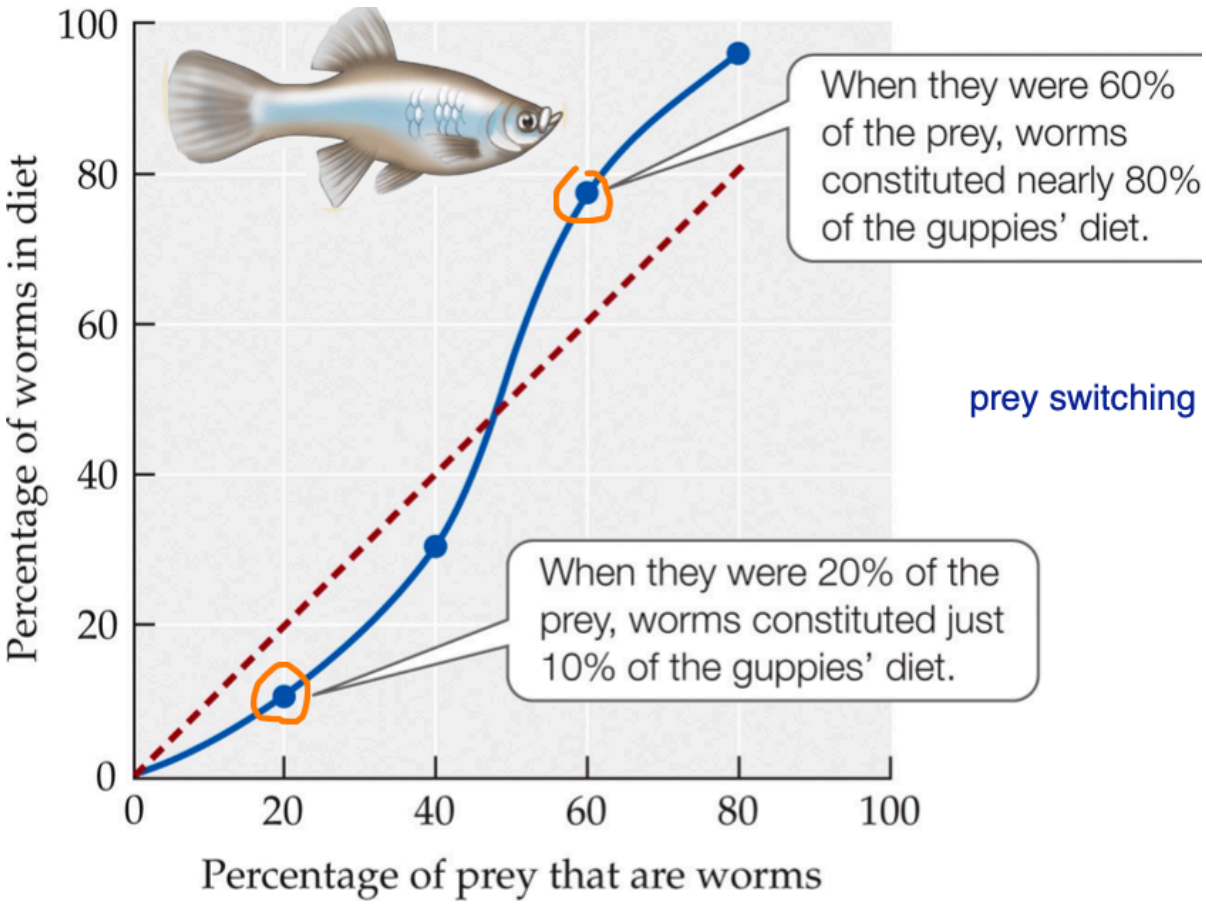
1. locate prey —--- crypsis, habitat selection

2. catch prey —--- speed / defensive behavior
3. consume prey —--- morphology/chemistry
4. digest prey —--- chemistry

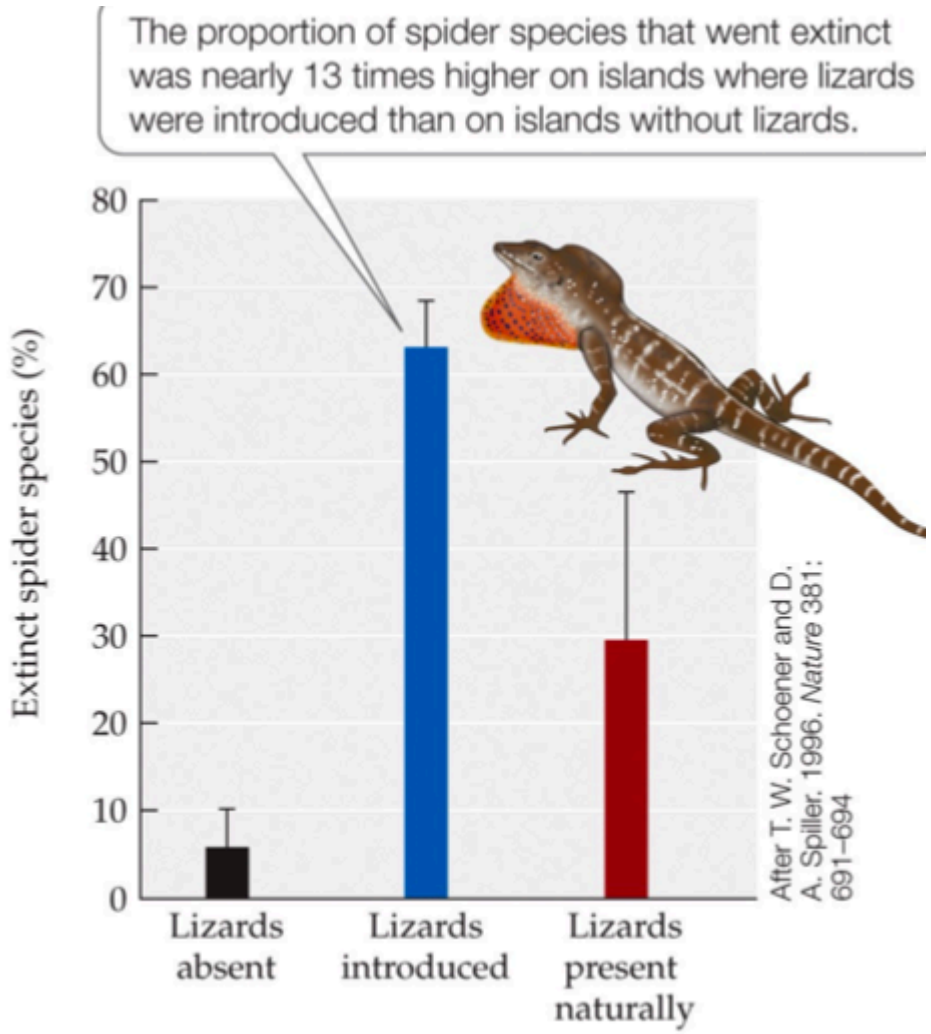
- How to avoid detection: crypsis eg. Snowshoe hare fur color changes
- How to avoid capture: be inaccessible eg. bark beetle larvae hide in wood
- How to avoid capture: be a bad ass eg. skunk
- How to avoid being eaten: morphological defenses eg. Elephant tusks
- How to avoid being eaten / digested: chemical defenses eg. Poisonous tree frog

E. Ecological importance of predation

- predators can control where prey live (fundamental vs. realized niche)
- predators can exert density-dependent control over their prey



- predators can drive their prey extinct…



- ...and they can also disrupt competitive exclusion
    - Predators (Pisaster) preferentially eat mussels and larger barnacles, preventing them from competitively excluding Chthamalus
- predators can strongly influence local biodiversity
    - Snow geese eat basically all grass

19. Parasitism

A. Key Concepts

- Parasites typically feed on only one or a few host species, but host species have multiple parasite species
- Hosts and parasites can have important effects on each other's population dynamics

- Parasites can alter the outcomes of species interactions, thereby causing communities to change
- **Symbiosis**: a relationship in which two species (symbionts) live in close physical and/or physiological contact

B. Parasite Natural History

- What range of species are commonly and unequivocally parasitic?
    - Protozoans: amoebae, ciliates, sporozoa
    - Helminths: nematode, cestode, trematode
    - Arthropods: insects, arachnids
    - Fungi
    - Bacteria
    - Virus

C. Endoparasitism and ectoparasitism

	Endoparasitism	Ectoparasitism
Advantages	Ease of dispersal	Ease of feeding
	Safe from host's immune system	Protected from external environment
		Safer from natural enemies
disadvantages	Vulnerability to natural selection	Vulnerability to host's immune system
	Exposure to external environment	Dispersal more difficult
	Feeding more difficult	

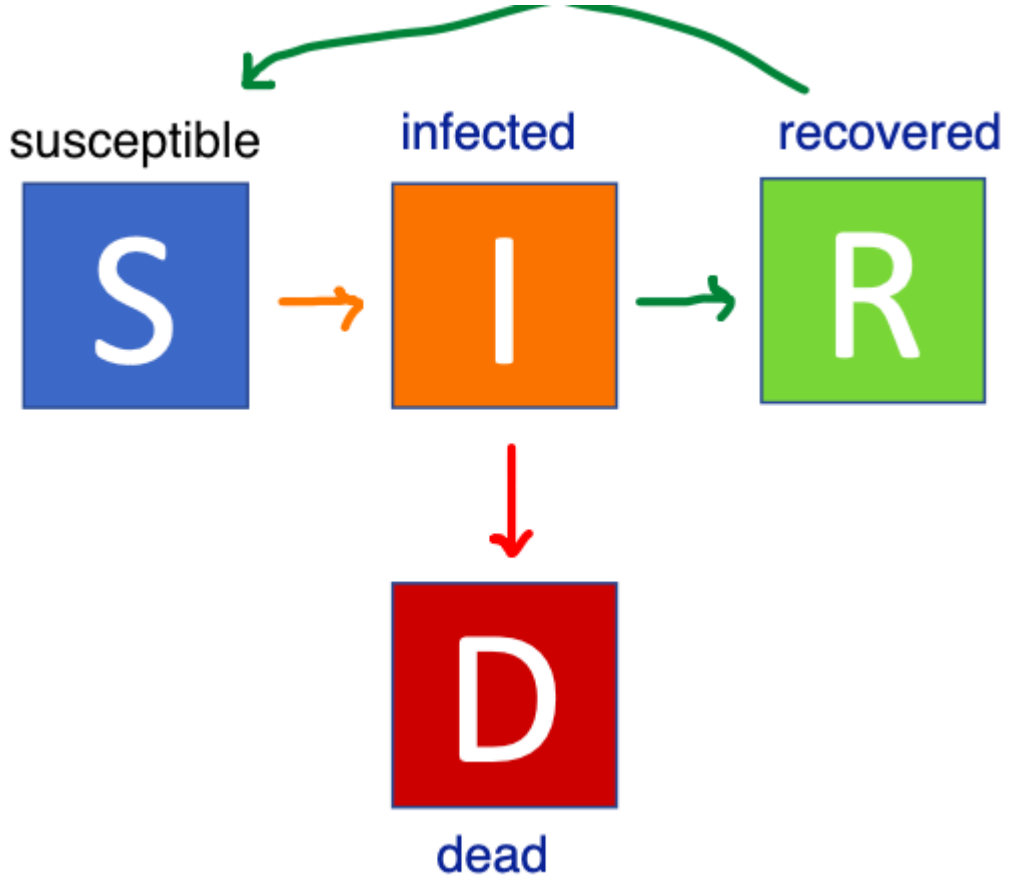
Solutions for endoparasites for dispersal

- Enslave host
    - Some fungi 'enslave' their insect hosts, altering the hosts' perching behaviour so that the fungal spores can be dispersed more easily after the host dies.
    - snails vs. the flatworm Leucochloridium

- Diarrhea
- Specialized life stages
    - Mistletoe let the birds eat the seeds and poop on other hosts
- More than 50% of all species can be classified as symbionts

D. host-pathogen dynamics

- Pathogen: a parasite that causes disease



$$\frac{dI}{dt} = \beta SI - mI$$

The change in the density of infected hosts depends on…

the number of susceptible hosts that become infected…

[a factor of how often susceptible hosts encounter an infected host times the probability of transmission]

minus the number of infected hosts that are no longer infected.

[equal to the number that die plus the number that recover]

dI/dt = the change in the density of infected individuals through time

S = the density of susceptible individuals

I = the density of infected individuals

β = the transmission coefficient (encounters that end in transmission)

m = the combined host death and recovery rate

- A disease will become established when the density of infected individuals in a population increases over time, which occurs when dI/dt > 0

$$dI/dt = \beta SI - mI$$

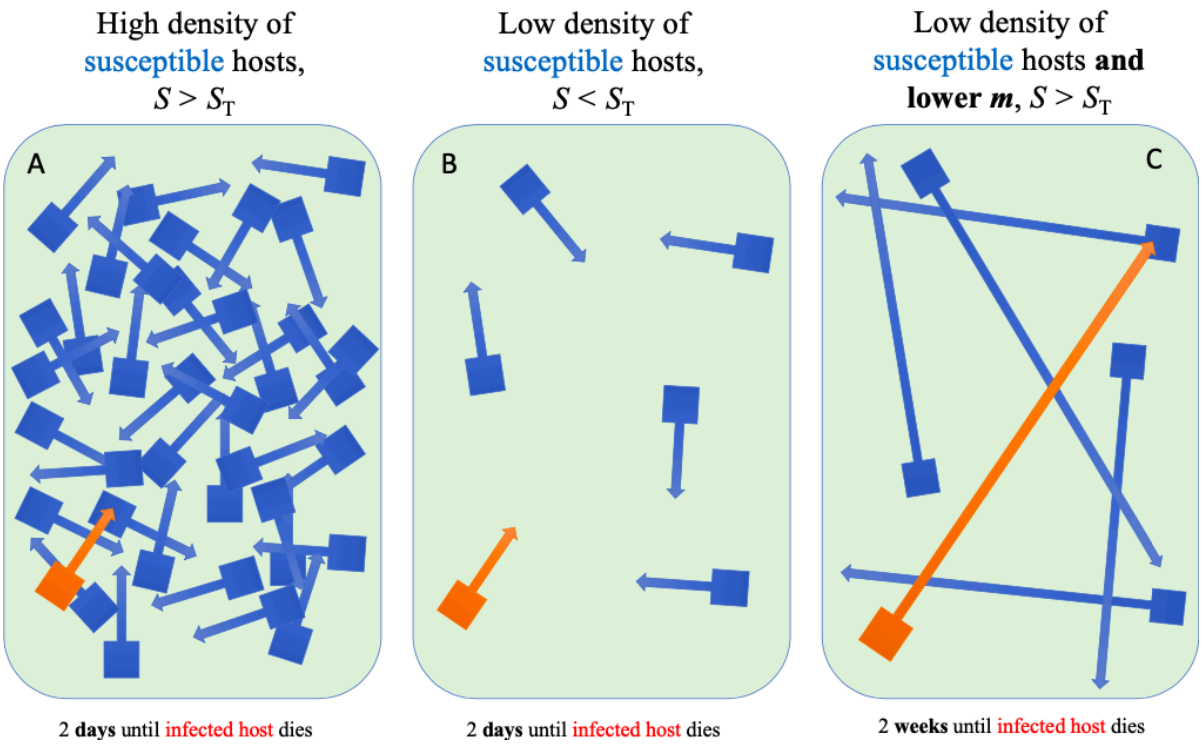
↳ > 0 disease will spread

which we can express as: $\beta SI - mI > 0$

or rearranged as: $S > m/\beta$

such that the "threshold density" is: $S_T = m/\beta$

- To manage the disease, we want S_T to be higher (m↑β↓)
    - We can attempt to keep the population belowS
    - We could either reduce the number of susceptible individuals through either vaccination or culling (I and also m)



20. Facilitation

A. Key Concepts

- Facilitation: In positive interactions, no species is harmed, and the benefits are greater than the costs for at least one species (+/+ or +/0)
- Each partner in a mutualistic interaction acts in ways that serve its own ecological and evolutionary interests
- Positive interactions affect the abundances and distributions of populations as well as the structure of ecological communities

B. Positive interactions

- Positive interactions (a.k.a facilitation) are those interactions in which one or both species benefit and neither is harmed
    - Benefits of positive interactions can take many forms— food, shelter, transport, etc.

- Sometimes there is a cost to one or both partners, but the net effect is positive for at least one partner.

Facilitation includes:

- Mutualism: Mutually beneficial interaction between individuals of two species (+/+ relationship).
    - Most plants form associations with fungi that improve the plant's growth and survival and probably have done this for 400 million years.
    - Most flowering plants depend on pollinators, and long-term increases in insect diversity are thought to result from this relationship
    - Endosymbiotic gut microbes are critical for animals that consume plant matter because they can break down cellulose
    - Photosynthetic endosymbionts enhance the success of corals and sea anemones
- Commensalism: Individuals of one species benefit; individuals of the other species do not benefit but are not harmed (+/0 relationship).
    - Habitat providing
    - Lichens grow on trees.
    - Bacteria live on human skin.
    - In kelp forests, many species depend on the kelp for habitat and do no harm to the kelp.

C. Effect of Positive Interactions:

① Some positive interactions are highly species specific and obligate (not optional for either species).

- Example: Tropical figs are pollinated by fig wasps.

- The wasps and the figs have **coevolved**, and neither species can reproduce without the other.
- The wasps have complex reproductive behaviors in the fig receptacle that ensure pollination. Wasp larvae develop by eating some of the fig seeds.

② The strength of positive interactions can vary in space and time

- Relative neighbor effect (RNE) = target species' growth with neighbors present minus its growth with neighbors removed.
- RNE was generally **positive at high- elevation** sites: neighbors had a positive effect on the target species.
- RNE was generally **negative at low- elevation** sites.
- RNE was negatively related to maximum summer temperature, suggesting that **positive interactions were more common in colder, more stressful environments** and **competition was more common in warmer,** less stressful environments.

D. Mutualisms

① Trophic mutualisms:

- A mutualist receives energy or nutrients from its partner.

- Leaf-cutter ants and fungus
- Mycorrhizae

② Habitat mutualisms:

- One partner provides the other with shelter, living space, or favorable habitat.

- Pistol shrimp dig burrows that they share with goby fish. The goby gets a refuge and in turn serves as a "seeing eye fish" for the nearly blind shrimp.
- corals provide habitat for sea urchins, which graze algae that can otherwise overgrow the corals.

21. Communities I: disturbance and succession

A. Key Concepts

- Processes such as disturbance, stress, predation, and positive interactions can mediate resource availability, thus promoting species diversity
- Succession is the process of change in species composition over time as a result of abiotic and biotic agents of change

B. Cause of Death and Destruction

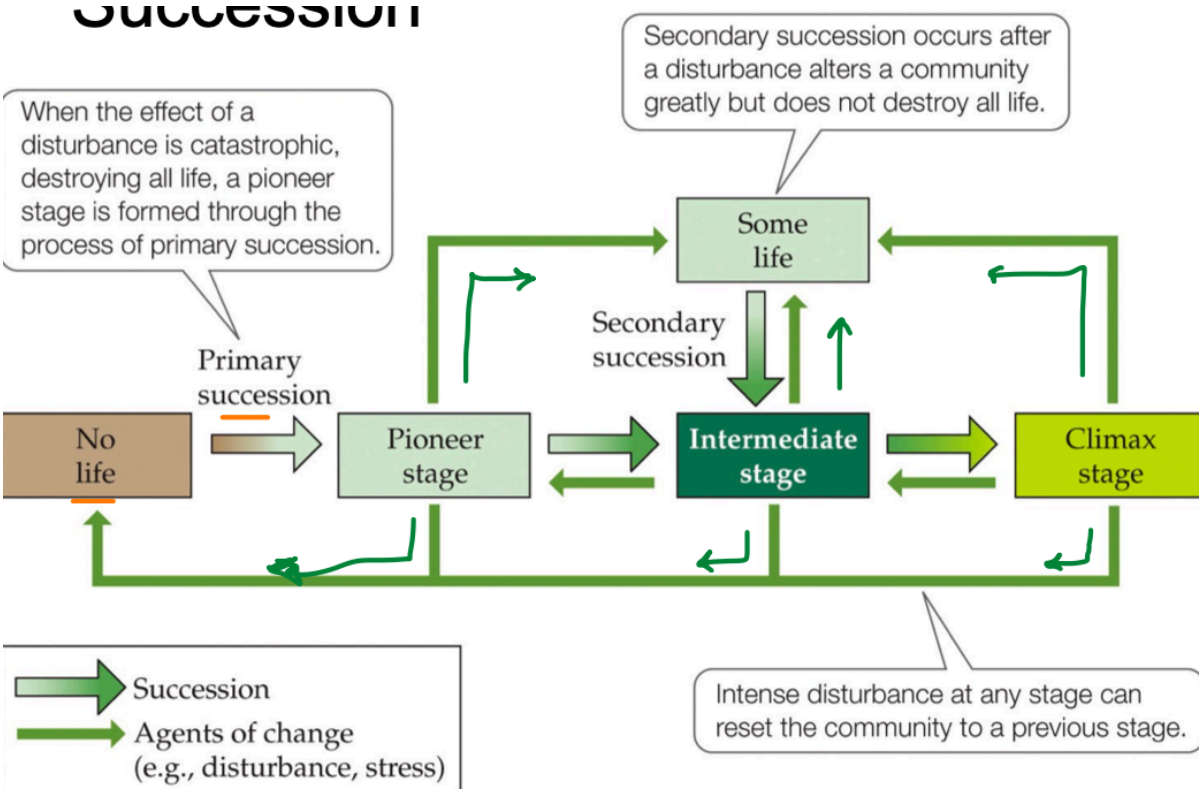
1. Stress (too hot, too cold, too salty, too dry, etc.)
2. Species interactions (e.g., competition, predation)
3. **Disturbance**: an **abiotic** event that physically injures or kills some individuals and creates opportunities for other individuals to grow or reproduce.

C. Succession

- Change in the species composition of communities over time as a result of a variety of abiotic (physical and chemical) and biotic agents of change.
- The non-seasonal, directional, and continuous pattern of colonization and extinction on a site by species populations.

Primary vs. Secondary Succession:

- **primary succession** starts from scratch, and occurs when new substrate becomes available for the first time (e.g. lava flows, rock fracture)
- **secondary succession** follows a disturbance that leaves some components of a community (e.g. spores, microscopic organisms) in place



Models of Succession

- FACILITATION: early successional species speed succession by improving conditions for later colonists
- INHIBITION: early successional species delay succession by interfering with (e.g., competing with) later colonists
- TOLERANCE: early successional species have neither a positive nor negative influence on later colonists

Traits of early successional species, and

Traits of of species in highly disturbed habitats:

- r selected
    - good dispersers
    - fast growers
    - early maturity
    - high reproductive rates

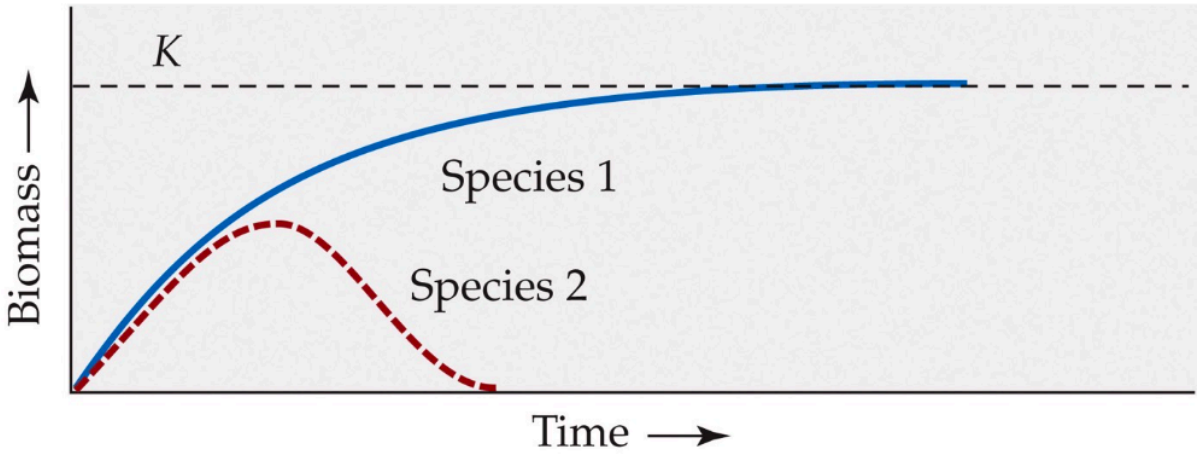
Traits of late successional species, and

Traits of of species in rarely disturbed habitats:

- K selected
    - good competitors
    - slower growers
    - later maturity
    - Long-lived

D. Competition in variable conditions

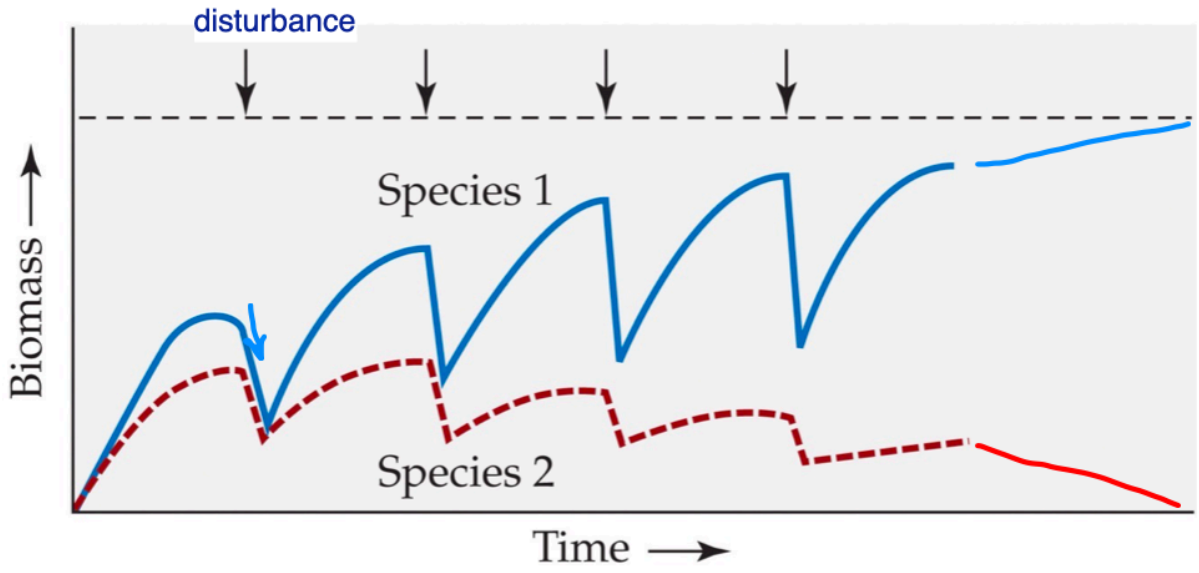
a. Constant condition



b. Variable condition

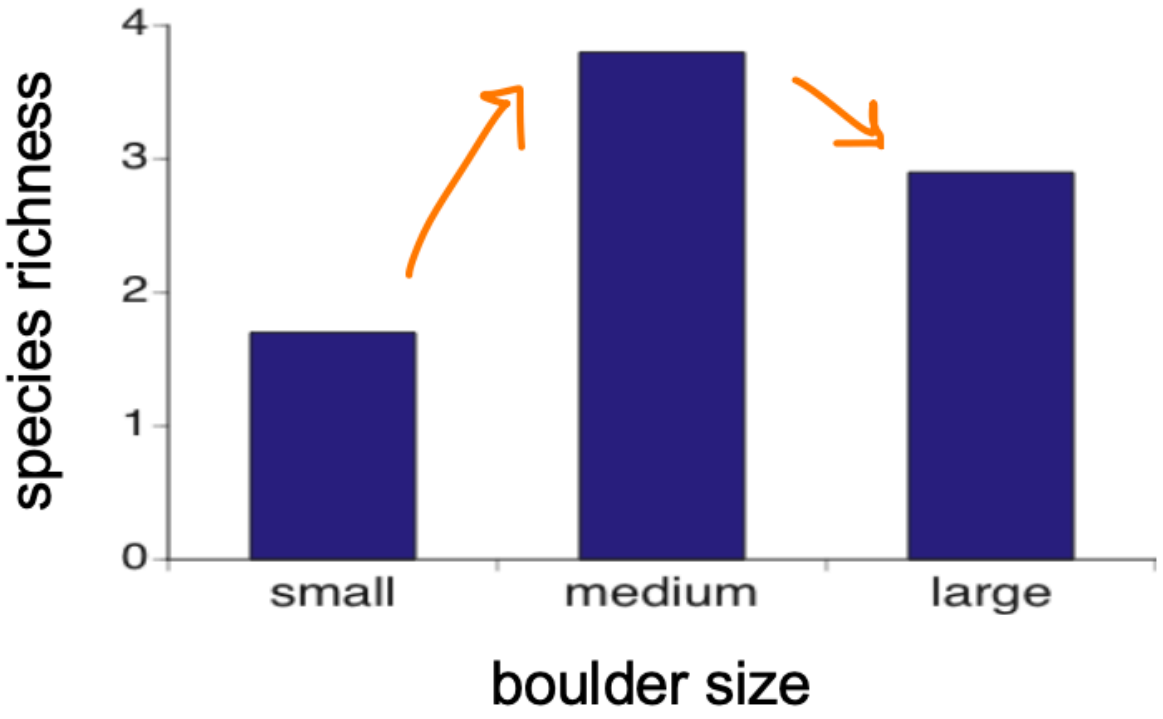
i. Environmental variability can prevent the competitive dominant from excluding the competitively inferior species.

ii. Populations begin to grow again, competition and other interactions become important, and with enough time the community may begin to look like the pre-disturbance community.

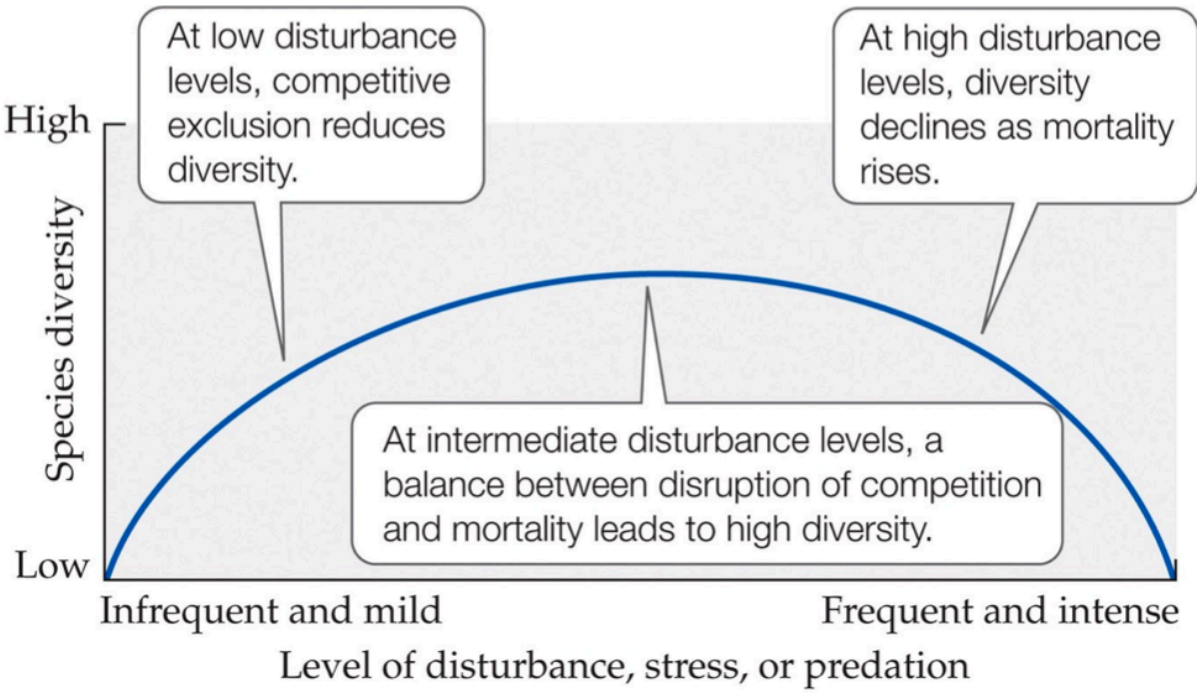


E. Case Study: In intertidal boulder fields, small boulders are disturbed (rolled by waves) more frequently than large boulders

- High disturbance on small boulders means that only a few species can persist
    - r-selected species dominate
- Intermediate disturbance on medium boulders allows a mix of early and late successional species to coexist
- Low disturbance on large boulders means that early successional species have been competitively excluded
    - K-selected species dominate



F. Intermediate Disturbance Hypothesis



Diversity peaks at intermediate levels of disturbance when:

1. competitively inferior species are better at colonizing new patches (ie r-selected), or...
2. competitively inferior species are more resistant to the disturbance

Case Study:

- Darwin was the first to consider disturbance as a mechanism to maintain species diversity.
- In a meadow that he stopped mowing, he observed that the species number went from 20 down to 11.
- Without disturbance (mowing), the dominant species were able to exclude several others.

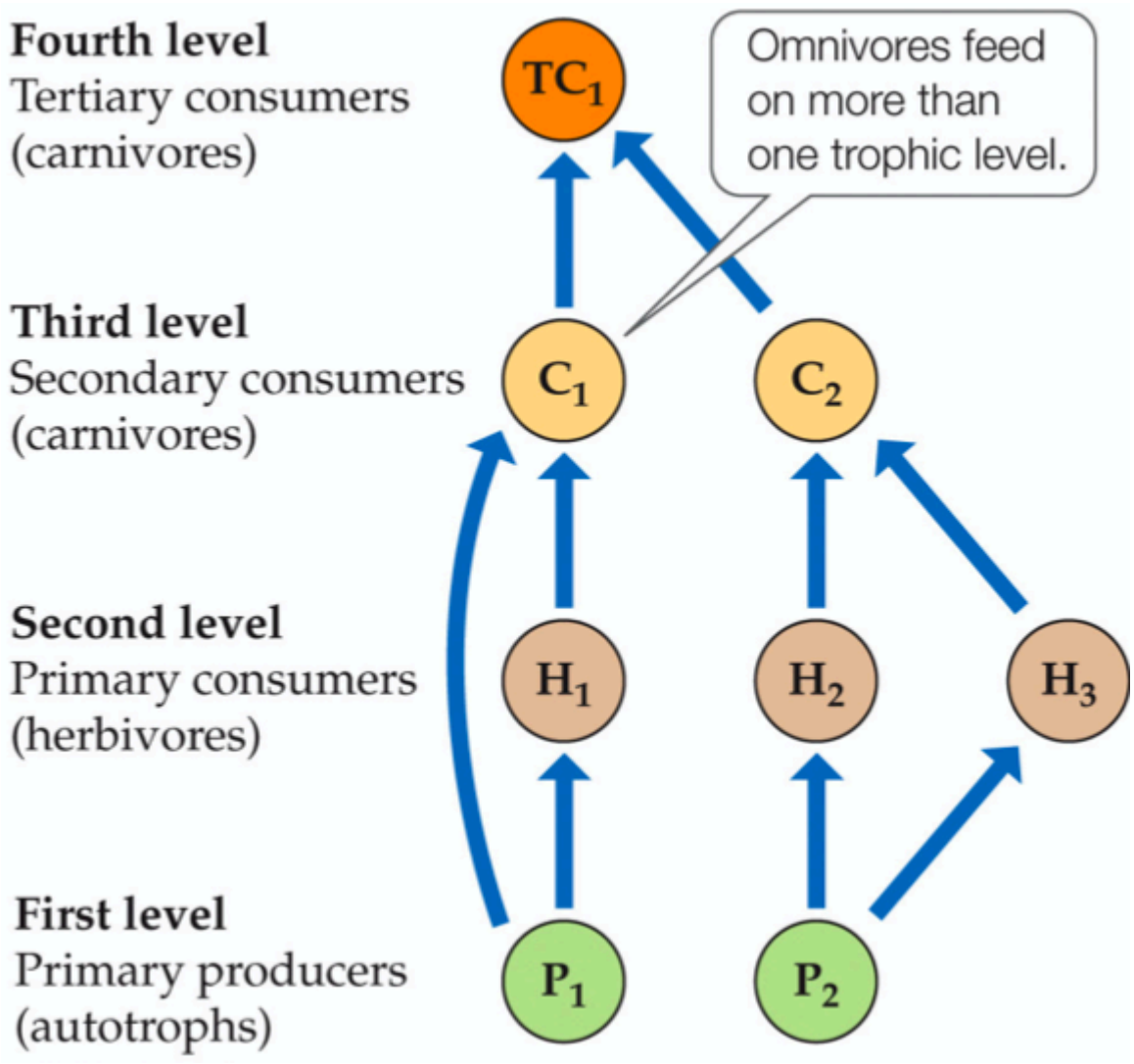
22. Communities II: Interaction webs

A. Key Concepts

- Communities can be characterized by complex networks of direct and indirect interactions that vary in strength and direction

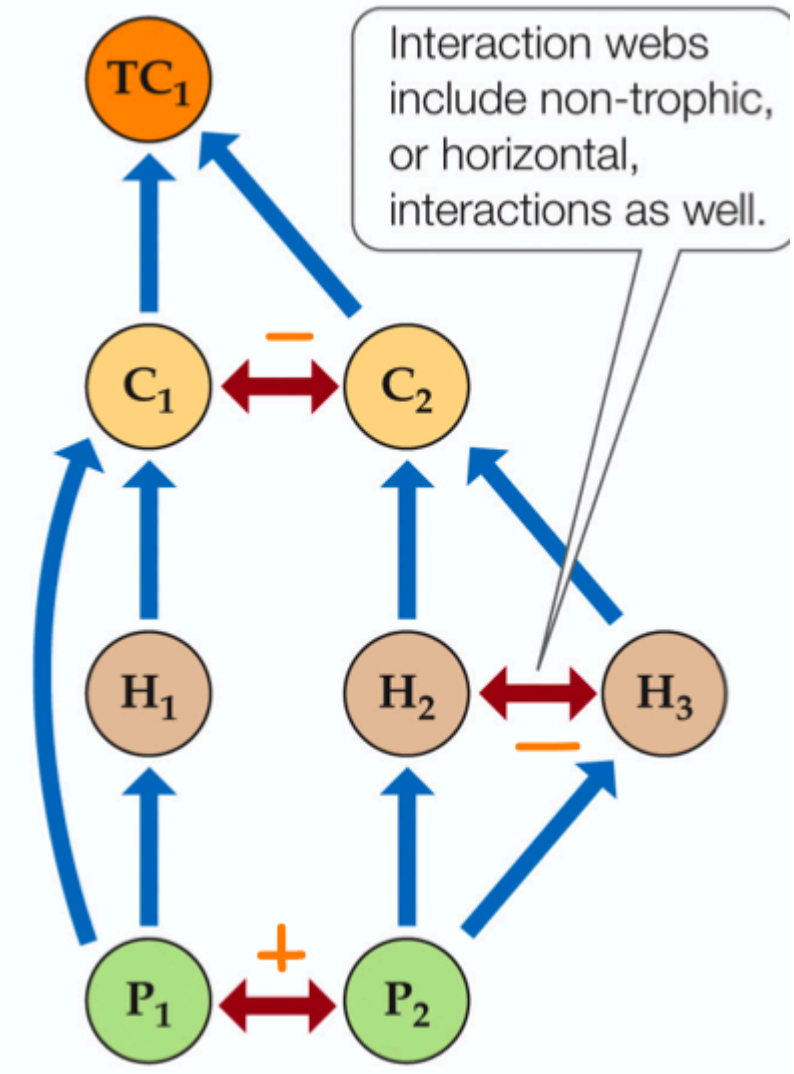
B. Food web

- Just trophic interactions

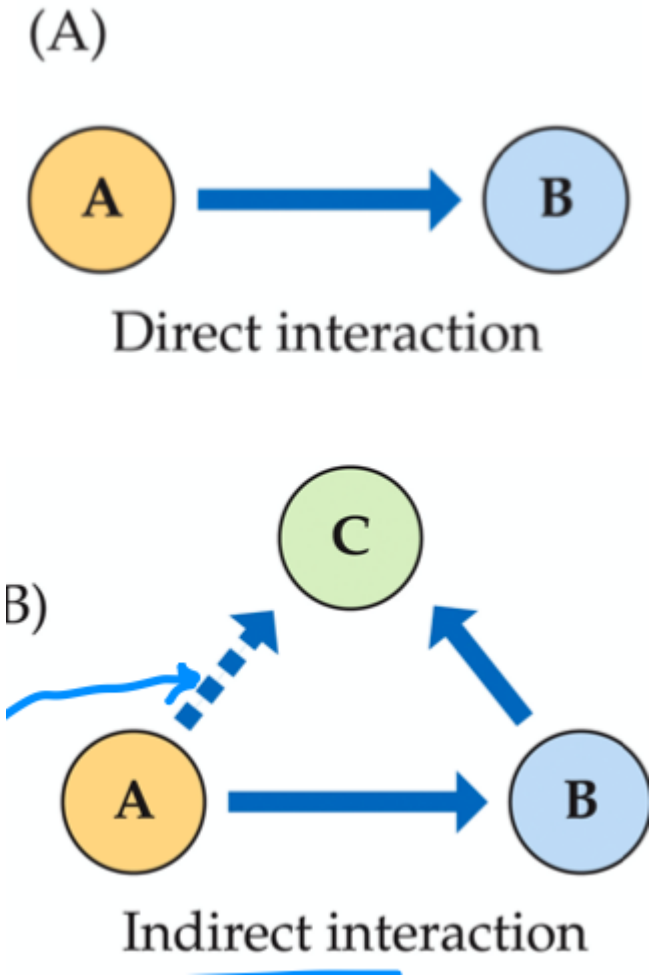


C. Interaction web

- All interspecific interactions

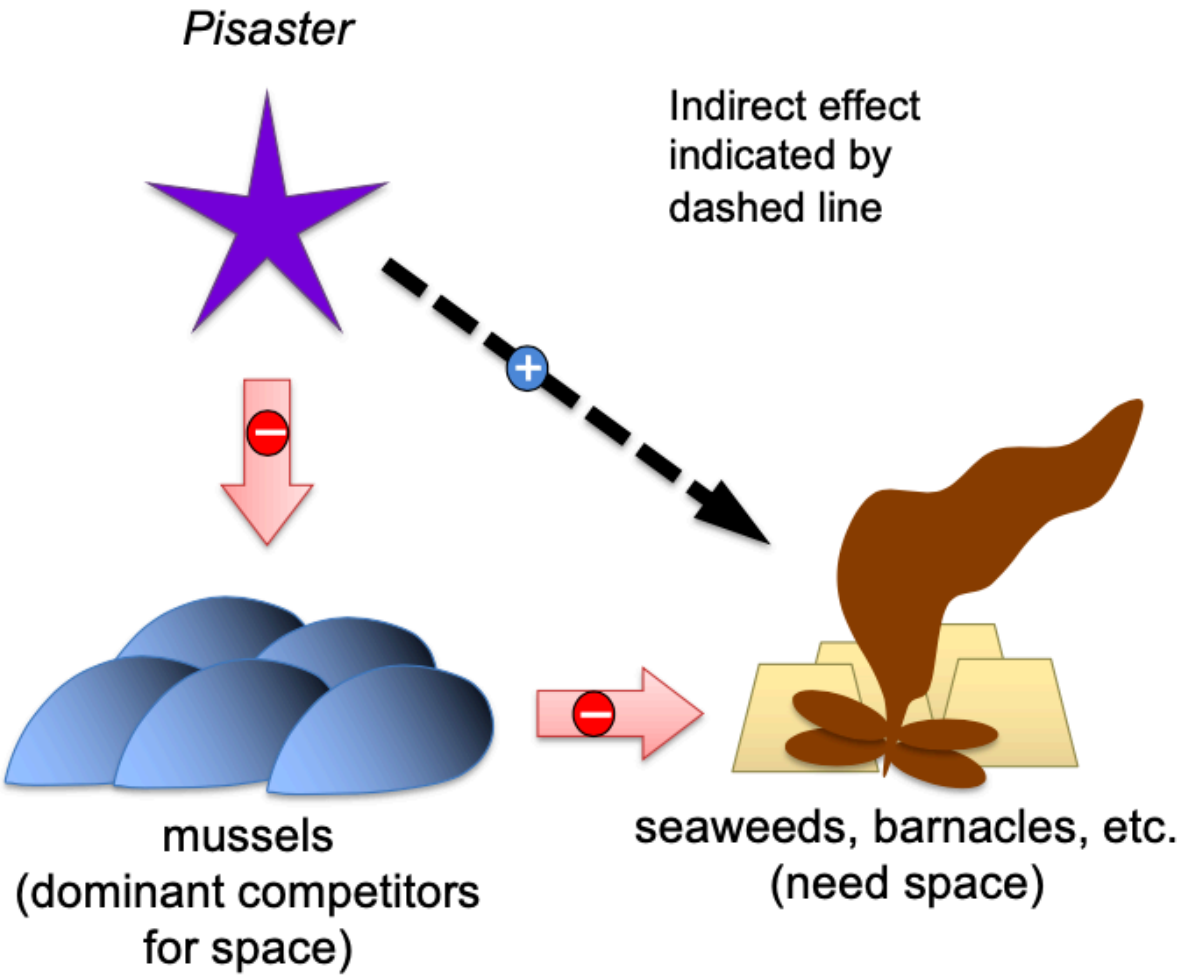


- Direct interactions: Between two species (e.g., competition, predation, facilitation).
- Indirect interactions: The relationship between two species is mediated by at least one additional species



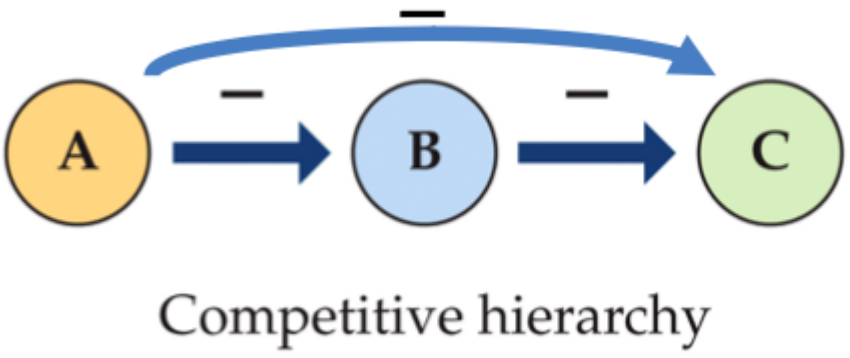
D. How do indirect effects influence diversity?

① Pisaster has an indirect, positive effect on several species by preventing mussels from monopolizing space, and species richness of space-requiring species increases from 8 to 15 when Pisaster is present



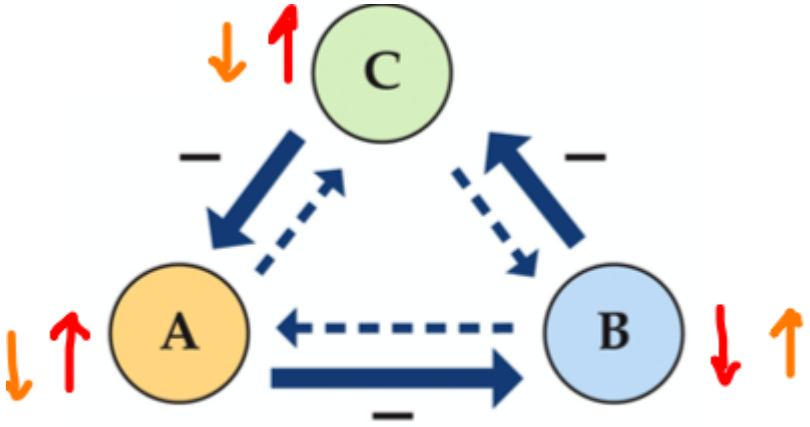
② Competitive networks:

- Interactions among multiple species in which no species wins every pair-wise competitive interaction.
- In a network, as opposed to a linear hierarchical system, no one species dominates the interaction, allowing for coexistence.



Competitive hierarchy

- In a competitive hierarchy, A outcompetes B, but this does not benefit C because A also outcompetes C.

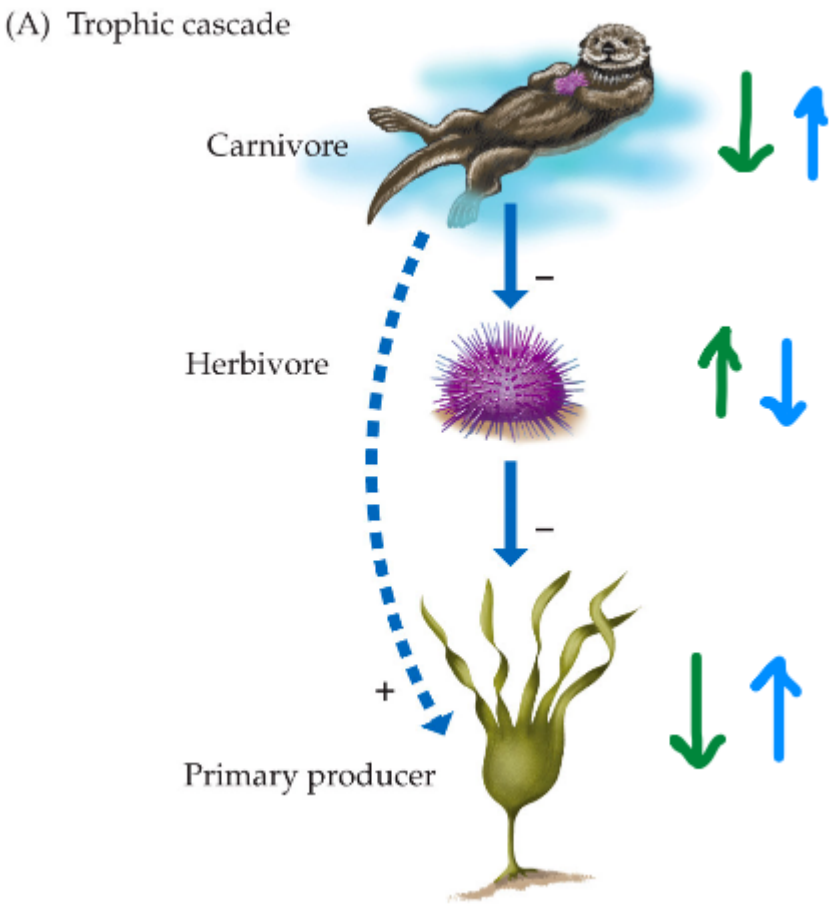


Competitive network

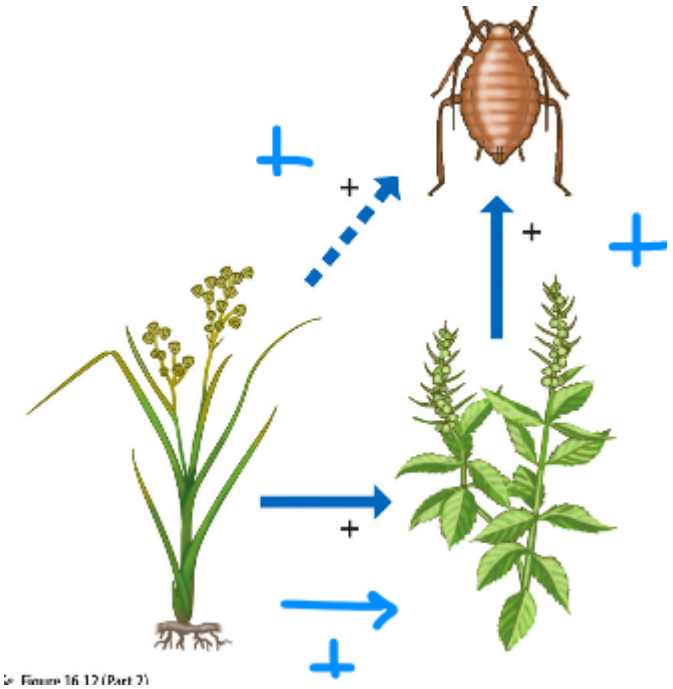
- In a competitive network, A outcompetes B, which benefits C, which outcompetes A, which benefits B, which outcompetes C, which benefits A, which....

③ Importantance

- Trophic cascade:
    - A carnivore eats an herbivore (a direct negative effect on the herbivore).
    - The decrease in herbivore abundance has a positive effect on a primary producer.



- Trophic facilitation:
    - A consumer is indirectly facilitated by a positive interaction between its prey and another species.
    - Example: Interactions between salt marsh plants affect aphids



- Indirect interactions can be generated by changes in species density and/or by changes in species traits (including behavioural traits)
    - A density-mediated indirect interaction (DMII) occurs when species A affects the density of species B, with consequences for species C (e.g., a classical trophic cascade)
    - A trait-mediated indirect interaction (TMII), sometimes known as a behaviourally-mediated indirect interaction (BMII), occurs when species A affects the behaviour (or some other trait) of species B, with consequences for species C (e.g., a landscape of fear scenario)
        - Wild donkeys are avoiding valleys frequented by cougars, leading to excessive growth of local shrubs.
        - The survival rate of hummingbirds near the eagle's nest has increased because Jay, a predator of hummingbirds, avoids it.

E. Terminology

① Keystone species

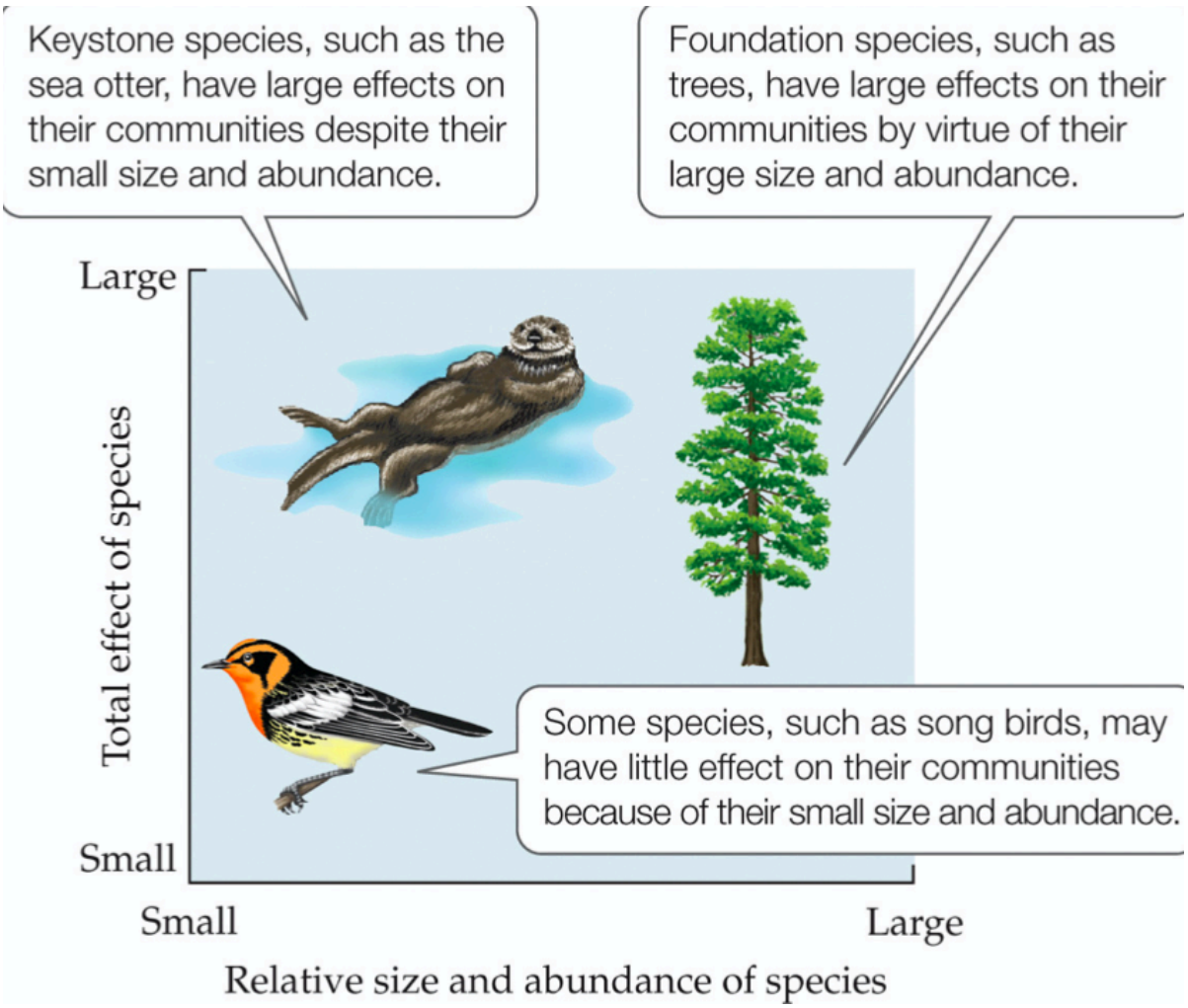
- have strong effects because of their role in a community.
- Their effect is large in proportion to their biomass or abundance.
- They usually influence community structure indirectly, generally via trophic means, as in the case of sea otters.

② Ecosystem engineers

- create, modify, or maintain physical habitat
- Example: Beavers build dams, which transform a swiftly flowing stream into a pond or marsh with wetland plants.
- Simply providing physical structure, like trees or corals, is another form of ecosystem engineering

③ Foundation species

- provide habitat [or food]
- Usually include those species actually contributing most of the spatial structure of the community, the competitive dominants
- Examples: trees, kelps, corals


Keystone species, such as the sea otter, have large effects on their communities despite their small size and abundance.
Foundation species, such as trees, have large effects on their communities by virtue of their large size and abundance.
Some species, such as song birds, may have little effect on their communities because of their small size and abundance.
Large
Small
Total effect of species
Small
Large
Relative size and abundance of species